
Theoretical and Practical Analysis of Fréchet Regression via Comparison Geometry

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Abstract

Fréchet regression extends classical regression methods to non-Euclidean metric spaces, enabling the analysis of data relationships on complex structures such as manifolds and graphs. This work establishes a rigorous theoretical analysis for Fréchet regression through the lens of comparison geometry which leads to important considerations for its use in practice. The analysis provides key results on the existence, uniqueness, and stability of the Fréchet mean, along with statistical guarantees for nonparametric regression, including exponential concentration bounds and convergence rates. Additionally, insights into angle stability reveal the interplay between curvature of the manifold and the behavior of the regression estimator in these non-Euclidean contexts. Empirical experiments validate the theoretical findings, demonstrating the effectiveness of proposed hyperbolic mappings, particularly for data with heteroscedasticity, and highlighting the practical usefulness of these results.

1 Introduction

Fréchet regression [35] is a powerful statistical tool for analyzing relationships between variables when the response or predictor lies in a non-Euclidean space. It generalizes classical regression to settings where the response variable Y resides in a metric space $\mathcal{M}$. Given predictors X , Fréchet regression seeks to estimate the conditional Fréchet mean.

$$\mu(x) = \arg \min_{m \in \mathcal{M}} \mathbb{E} [d^2(Y, m) \mid X = x], \quad (1)$$

where d is the metric on $\mathcal{M}$. This approach accommodates data in various non-Euclidean spaces, such as manifolds, trees, and graphs [29, 17, 18, 36, 13]. In recent years, several variants of Fréchet regression have been proposed [39, 7, 37, 19, 44, 42], each addressing different aspects such as variable selection, error modeling, and high-dimensional data handling. However, most existing studies primarily focus on specific geometric settings or lack a comprehensive theoretical framework that accounts for varying curvature bounds. This study fills this gap by leveraging comparison geometry to provide a unified theoretical analysis of Fréchet regression across $\text{CAT}(K)$ spaces with diverse curvature properties.

Fréchet regression allows the assumption of a non-Euclidean space in the space of the data, so one can expect that its behavior can be described depending on the geometrical properties of the space. To investigate this, this study utilizes comparison geometry, which is a fundamental branch of differential geometry that investigates the geometric properties of a given space by comparing it to model spaces of constant curvature [12, 20, 11, 41]. Unlike information geometry [3, 5, 33, 4, 27, 28], which focuses on general statistical manifolds, this framework leverages classical comparison theorems to derive insights about the structure and behavior of more complex or less regular spaces. By establishing inequalities and structural similarities between a target space and well-understood model

spaces (e.g., Euclidean, spherical, or hyperbolic geometries), comparison geometry enables the extension of geometric and topological results to broader contexts, including spaces that may lack smoothness or traditional manifold structures. In this framework, $\text{CAT}(K)$ spaces are pivotal objects of study, which are the generalization of constant curvature space [6, 22, 9]. $\text{CAT}(K)$ spaces are geodesic metric spaces, where geodesic triangles are thinner than their comparison triangles in the model space of constant curvature K . Consider several known examples of $\text{CAT}(K)$ spaces. Euclidean spaces $\mathbb{R}^n$ are classic examples with $K = 0$, exhibiting flat geometry. Hyperbolic spaces, which have constant negative curvature ($K < 0$), serve as models for spaces exhibiting exponential growth and are useful in areas like network analysis and evolutionary biology. On the other hand, trees can be viewed as $\text{CAT}(0)$ spaces, providing a discrete analog with unique geodesics between points. Additionally, certain types of manifold structures used in shape analysis and computer graphics also qualify as $\text{CAT}(K)$ spaces under specific curvature conditions. These examples demonstrate the broad applicability of $\text{CAT}(K)$ spaces in modeling diverse geometric contexts encountered in statistical analysis. By considering such spaces, this study aims to describe the behavior of the Fréchet regression in terms of curvature K in particular.

2 Notation

In this section, the notations and definitions required for the following analysis are organized. Let $\mathcal{M}$ be a metric space and d be the metric on $\mathcal{M}$. Here, the metric space $(\mathcal{M}, d)$ is geodesic space if every pair of points in $\mathcal{M}$ can be connected by a geodesic, a curve whose length equals the distance between the points.

Definition 1 ($\text{CAT}(K)$ space). Let $(\mathcal{M}, d)$ be a geodesic metric space and let $K \in \mathbb{R}$. The space $\mathcal{M}$ is said to be a $\text{CAT}(K)$ space if it satisfies the following curvature condition: for any geodesic triangle $\triangle pqr$ in $\mathcal{M}$ with perimeter less than $2D_K$ (where $D_K = \pi/\sqrt{K}$ if $K > 0$ and $D_K = \infty$ otherwise), and for any points x, y on the edges $[pq]$ and $[qr]$ respectively, the distance between x and y in $\mathcal{M}$ does not exceed the distance between the corresponding points $\bar{x}$ and $\bar{y}$ on the comparison triangle $\triangle p\bar{q}\bar{r}$ in the model space of constant curvature K : $d(x, y) \leq d_{\mathbb{M}_K^2}(\bar{x}, \bar{y})$, where the comparison triangle $\triangle p\bar{q}\bar{r}$ is a triangle in the simply connected, complete 2-dimensional Riemannian manifold $\mathbb{M}_K^2$ of constant curvature K that preserves the side lengths as $d_{\mathbb{M}_K^2}(\bar{p}, \bar{q}) = d(p, q)$, $d_{\mathbb{M}_K^2}(\bar{q}, \bar{r}) = d(q, r)$, and $d_{\mathbb{M}_K^2}(\bar{r}, \bar{p}) = d(r, p)$.

Definition 2 (Geodesic convexity). A function $f: \mathcal{M} \rightarrow \mathbb{R}$ is geodesically convex if for every geodesic $\gamma: [0, 1] \rightarrow \mathcal{M}$, $f(\gamma(t)) \leq (1-t)f(\gamma(0)) + tf(\gamma(1))$ for all $t \in [0, 1]$.

Definition 3 (λ -strong geodesic convexity). A function $f: \mathcal{M} \rightarrow \mathbb{R}$ is λ -strongly geodesically convex around $p \in \mathcal{M}$ if there exists a constant $\lambda > 0$ depending only on K and $\text{diam}(\mathcal{M})$ such that

$$f(x) - f(p) \geq \lambda d^2(x, p), \quad (2)$$

for every $x \in \mathcal{M}$.

Definition 4 (Lower semicontinuity). A functional $F: \mathcal{M} \rightarrow \mathbb{R} \cup \{+\infty\}$ is lower semicontinuous at a point $x \in \mathcal{M}$ if for every sequence $\{x_n\}$ converging to x , it satisfies

$$F(x) \leq \liminf_{n \rightarrow +\infty} F(x_n). \quad (3)$$

Definition 5 (Weak convergence in metric space). A sequence of probability measures $\{\nu_n\}$ on $\mathcal{M}$ is said to converge weakly to a probability measure ν (denoted by $\nu_n \Rightarrow \nu$) if for every bounded continuous function $f: \mathcal{M} \rightarrow \mathbb{R}$,

$$\lim_{n \rightarrow +\infty} \int_{\mathcal{M}} f(y) d\nu_n(y) = \int_{\mathcal{M}} f(y) d\nu(y).$$

Definition 6 (Alexandrov angle). The Alexandrov angle $\angle_x(y, z)$ is defined as the limit of secular angles between short sub-segments. Concretely, if y' is a point on $[xy]$ with $d(x, y') \rightarrow 0$ and z' is a point on $[xz]$ with $d(x, z') \rightarrow 0$. Then,

$$\angle_x(y, z) := \lim_{y' \rightarrow x, z' \rightarrow x} \angle_x^{(\text{sec})}(y'z').$$

where $\angle_x^{(\text{sec})}(y'z')$ is the ordinary angle in the comparison triangle for $\triangle xy'z'$ in the model space.

Definition 7 (Riemannian exponential map). Let $T_z\mathcal{M}$ be the tangent space of $\mathcal{M}$ at a point $z \in \mathcal{M}$. For a fixed point z , the Riemannian exponential map at z , denoted by $\exp_z$, is a map from the tangent space at z to the manifold $\mathcal{M}$: $\exp_z: T_z\mathcal{M} \rightarrow \mathcal{M}$. Here, the Riemannian exponential map is constructed as i) Choose a tangent vector $v \in T_z\mathcal{M}$. ii) Consider the unique geodesic $\gamma_v(t)$ emanating from z with initial velocity v . Formally, $\gamma_v(t)$ satisfies $\gamma_v(0) = z$ and $\gamma'_v(0) = v$. iii) The exponential map sends the tangent vector v to the point on the manifold reached by traveling along the geodesic γ_v for unit time, $\exp_z(v) = \gamma_v(1)$.

3 Theory

See Appendix B for complete proofs of all statements.

3.1 Key Lemmas

Here, we summarize key lemmas required for our study. These results follow those of previous studies [43, 23, 24], but are presented below for the sake of uniformity of notation and to keep the manuscript self-contained. First, it can be shown that in $\text{CAT}(K)$ spaces with $K < 0$, the convexity properties ensure the existence and uniqueness of the Fréchet mean under mild conditions. For $\text{CAT}(K)$ spaces with $K > 0$, additional constraints on the diameter of the space may be necessary to ensure uniqueness due to potential multiple minima arising from positive curvature.

Lemma 1. Let $(\mathcal{M}, d)$ be a $\text{CAT}(K)$ space for $K < 0$. For any fixed point $p \in \mathcal{M}$, the function $f: \mathcal{M} \rightarrow \mathbb{R}$ defined by $f(x) = d^2(p, x)$ is geodesically convex.

Lemma 1 establishes that the squared distance function retains geodesic convexity in $\text{CAT}(K)$ spaces with non-positive curvature. This property is fundamental because it ensures that the Fréchet functional, which aggregates squared distances, inherits convexity. Consequently, optimization procedures to find the Fréchet mean are well-behaved, avoiding local minima and guaranteeing global optimality under the given conditions.

Lemma 2. Let $(\mathcal{M}, d)$ be a complete $\text{CAT}(K)$ space. For any probability measure ν on $\mathcal{M}$ with compact support, there exists at least one minimizer $m \in \mathcal{M}$ of the Fréchet functional:

$$m = \arg \min_{x \in \mathcal{M}} \int_{\mathcal{M}} d^2(y, x) d\nu(y).$$

Lemma 3. Let $(\mathcal{M}, d)$ be a $\text{CAT}(K)$ space with $K < 0$ that is strictly geodesically convex, meaning that the squared distance function $f(x) = d^2(p, x)$ is strictly geodesically convex for any fixed point $p \in \mathcal{M}$. Then, for any probability measure ν on $\mathcal{M}$ with compact support, the Fréchet mean m is unique.

Based on Lemma 1, which ensures geodesic convexity of the squared distance function in non-positively curved $\text{CAT}(K)$ spaces, and Lemma 2, which guarantees the existence of a Fréchet mean under compact support, one can establish the stability of the Fréchet mean under measure perturbations. Furthermore, Lemma 3 ensures uniqueness under strict geodesic convexity, thereby enabling Proposition 1 to assert the convergence of Fréchet means in non-positively curved spaces.

Proposition 1. Let $(\mathcal{M}, d)$ be a $\text{CAT}(K)$ space with $K < 0$. Suppose $\{\nu_n\}$ is a sequence of probability measures on $\mathcal{M}$ that converges weakly to a probability measure ν . Assume that for each n , the measure ν_n has a unique Fréchet mean m_n , and ν also has a unique Fréchet mean m . Then, the sequence of Fréchet means $\{m_n\}$ converges to $m \in \mathcal{M}$.

Proposition 1 claims that the $\text{CAT}(K)$ condition with $K < 0$ ensures that the space is non-positively curved, which imbues the space with strict convexity properties crucial for the uniqueness and stability of minimizers. This geometric structure prevents the existence of multiple local minima, thereby facilitating the continuity of minimizers under perturbations of the measure. Here, the stability of the Fréchet mean under measure perturbations is foundational for Fréchet regression. It ensures that as predictors vary and induce changes in the conditional distributions of responses, the conditional Fréchet means (regression estimates) behave predictably and converge appropriately as sample size increases.

Lemma 4. Let $(\mathcal{M}, d)$ be a $\text{CAT}(K)$ space with positive curvature bound $K > 0$. If the diameter of the support of the probability measure ν , denoted by $\text{diam}(\text{supp}(\nu))$, satisfies $\text{diam}(\text{supp}(\nu)) < \frac{\pi}{2\sqrt{K}}$, then the Fréchet mean m of ν is unique.

In Lemma 4, the diameter constraint ensures that all points in the support of ν lie within a geodesic ball of radius $R = \pi/2\sqrt{K}$. In $\text{CAT}(K)$ spaces with $K > 0$, such balls are geodesically convex, meaning any geodesic between two points within the ball lies entirely inside the ball. This local convexity is crucial for preserving strict convexity properties of the Fréchet functional.

In addition, applying Lemmas 2 and 3, the following statement can be obtained.

Lemma 5. Let $(\mathcal{M}, d)$ be a complete $\text{CAT}(K)$ space and consider a conditional distribution ν_x of Y given $X = x$. If for each x , the support of ν_x satisfies

$$\text{diam}(\text{supp}(\nu_x)) < D_K = \begin{cases} +\infty & \text{if } K \leq 0, \\ \frac{\pi}{\sqrt{K}} & \text{if } K > 0, \end{cases}$$

then then the conditional Fréchet mean in Eq. (1) exists and is unique for each x .

3.2 Convergence Rates and Concentration

Let $\hat{\mu}^*$ denote a nonparametric Fréchet regression estimator (e.g., Nadaraya–Watson–type kernel smoothing [32, 40, 8] on the predictor space). Then, the following statements for the concentration results, the pointwise consistency, and rates of convergence can be obtained. The important point is that one has to rely on exponential concentration inequalities valid in $\text{CAT}(K)$ spaces (e.g., specific versions of concentration of measure or deviation bounds for Fréchet means).

Theorem 1 (Concentration for the sample Fréchet mean). Let $(\mathcal{M}, d)$ be a complete $\text{CAT}(K)$ space of diameter at most D . Suppose that $Y_1, Y_2, \dots, Y_n$ are independent and identically distributed random points in $\mathcal{M}$, and let μ and $\hat{\mu}_n$ be the population and sample Fréchet mean.

$$\begin{aligned} \mu &:= \arg \min_{z \in \mathcal{M}} \mathbb{E}[d^2(Y, z)], \\ \hat{\mu} &:= \arg \min_{z \in \mathcal{M}} \frac{1}{n} \sum_{i=1}^n d^2(Y_i, z). \end{aligned}$$

Assume further that each $d^2(Y_i, z)$ is essentially bounded by D^2 or more generally that $d^2(Y_i, z)$ has sub-Gaussian tails uniformly in z . Then there exists $\delta > 0$ such that for every $\epsilon > 0$,

$$\mathbb{P}[d(\hat{\mu}, \mu) > \epsilon] \leq 2 \left(\frac{\alpha(K, D)D}{\delta} \right)^m e^{-\frac{n(\alpha(K, D)\epsilon^2)^2}{8D^2}}, \quad (4)$$

where m is the dimension of the manifold, and $\alpha(K, D)$ is the strong convexity constant.

In addition to the concentration for the sample Fréchet mean in the standard sense, the following proposition gives the concentration in L_p sense.

Proposition 2. Under the hypotheses of Theorem 1, there exist explicit constants $C_p(K, D)$ such that for any integer $n \geq 1$ and $p \geq 1$,

$$\mathbb{E}[d^p(\hat{\mu}_n, \mu)] \leq C_p(K, D)(n^{-p/2}). \quad (5)$$

That is, $d(\hat{\mu}_n, \mu)$ converges to 0 in L^p at a rate on the order of $n^{-p/2}$.

Moreover, the following theorem gives the pointwise consistency of nonparametric Fréchet regression in a $\text{CAT}(K)$ space. The main idea parallels classical kernel-based regression arguments in $\mathbb{R}^d$, but replaces ordinary arithmetic means by Fréchet means in the metric space $(\mathcal{M}, d)$.

Assumption 1 (Kernel LLN condition). For any bounded (or square-integrable) function $f: \mathcal{M} \rightarrow \mathbb{R}$ nonnegative weights $\{w_{n,i}(x)\}_{i=1}^n$ satisfies

$$\sum_{i=1}^n w_{n,i}(x) f(Y_i) \xrightarrow[n \rightarrow \infty]{a.s.} \mathbb{E}[f(x) \mid X = x]. \quad (6)$$

Theorem 2 (Pointwise consistency of nonparametric Fréchet regression). Let $\{(X_i, Y_i)\}_{i=1}^n$ be i.i.d. sample with $X_i \in \mathbb{R}^d$ and $Y_i \in \mathcal{M}$, where $(\mathcal{M}, d)$ is a complete $\text{CAT}(K)$ space with diameter $\text{diam}(\mathcal{M}) \leq D$. Define the population Fréchet regression function:

$$\mu^*(x) := \arg \min_{z \in \mathcal{M}} \mathbb{E}[d^2(Y, z) \mid X = x].$$

160 Assume that $\mu^*(x)$ is well-defined and unique for each x provided as Theorem 5. Also, let
 161 $\{w_{n,i}(x)\}_{i=1}^n$ be nonnegative weights that sum to 1 for each fixed x . For instance, in kernel re-
 162 gression, one sets

$$w_{n,i}(x) = \frac{W(\|x - X_i\|/h_n)}{\sum_{j=1}^n W(\|x - X_j\|/h_n)},$$

163 where $W(\cdot)$ is a usual kernel (with compact support or exponential decay), and $h_n \rightarrow 0$ is a
 164 bandwidth. Define the nonparametric Fréchet-regression estimator at x by

$$\hat{\mu}_n^*(x) = \arg \min_{z \in \mathcal{M}} \sum_{i=1}^n w_{n,i}(x) d^2(Y_i, z). \quad (7)$$

165 Then, under mild regularity conditions on the weights in Assumption 4, $\hat{\mu}_n^*(x) \xrightarrow[n \rightarrow \infty]{a.s.} \mu^*(x)$ for each
 166 fixed $x \in \mathbb{R}^d$.

167 Here, additional assumptions allow us to obtain the convergence rates in $\text{CAT}(K)$ spaces.

168 **Theorem 3** (Convergence rates in $\text{CAT}(K)$ spaces). Under the assumptions of Theorem 2, suppose
 169 additionally:

- 170 • $\mu^*: \mathbb{R}^d \rightarrow \mathcal{M}$ is β -Hölder (or Lipschitz) continuous, with respect to the usual Euclidean
 171 norm on $\mathbb{R}^d$ and the distance d on $\text{CAT}(K)$. That is, there exists $L > 0$ and $\beta > 0$ such
 172 that

$$d(\mu^*(x), \mu^*(x')) \leq L \cdot \|x - x'\|^\beta, \quad (8)$$

173 for all $x, x' \in \mathbb{R}^d$.

- 174 • The kernel weights $w_{n,i}(x)$ satisfy standard nonparametric conditions:

$$\sum_{i=1}^n w_{n,i}(x) = 1, \quad w_{n,i}(x) \approx W\left(\frac{\|x - X_i\|}{h_n}\right), \quad h_n \rightarrow 0, \quad nh_n^d \rightarrow +\infty. \quad (9)$$

- 175 • Each conditional distribution $Y \mid X = x$ has finite second moments in the $\text{CAT}(K)$ space
 176 and a unique Fréchet mean $\mu^*(x)$.
- 177 • The distribution of $Y \mid X = x$ varies smoothly in a local neighborhood of x . Formally, one
 178 assumes that for x' near x , the conditional distributions $\mathbb{P}[Y \in \cdot \mid X = x']$ do not differ too
 179 much, ensuring small bias when $x' \approx x$.

180 Then for the nonparametric Fréchet regression estimator $\hat{\mu}_n^*$

$$\sup_{x \in \mathcal{X}_0} \mathbb{E} [d^2(\hat{\mu}_n^*(x), \mu^*(x))] = O\left(\frac{1}{nh_n^d} + h_n^{2\beta}\right), \quad (10)$$

181 where $\mathcal{X}_0 \subseteq \mathbb{R}^d$ is any compact subset over which the kernel is applied.

182 From the above theorem, one can see that the usual $\left(\frac{1}{nh_n^d} + h_n^\beta\right)$ trade-off from Euclidean nonpara-
 183 metric statistics carries over to the $\text{CAT}(K)$ setting, once one accounts for i) geodesic convexity for
 184 controlling variance and ii) the Hölder continuity of $\mu^*(x)$ for controlling bias.

185 **Implications:** Section 3.2 provides the statistical properties of Fréchet regression estimators within
 186 $\text{CAT}(K)$ spaces. Theorem 1 offers exponential concentration bounds for the sample Fréchet mean,
 187 indicating that the estimator converges to the true mean with high probability as the sample size
 188 increases. Proposition 2 further quantifies this convergence in an L^2 sense, demonstrating that the
 189 expected distance between the sample and population Fréchet means decreases at a rate proportional to
 190 $n^{-1/2}$. These results are pivotal for understanding the efficiency and reliability of Fréchet regression
 191 estimators. They assure that given sufficient data, the regression estimates will not only be consistent
 192 but also achieve convergence rates comparable to those observed in classical Euclidean nonparametric
 193 regression.

3.3 Angle Stability for Conditional Fréchet Means

Understanding not just the position but also the directional relationships around the Fréchet mean is crucial for capturing the local geometry of the data distribution. Angle stability ensures that small perturbations in the underlying probability measures or data configurations do not lead to significant distortions in the angular relationships among points relative to the Fréchet mean. This property is particularly valuable when analyzing directional data or when the regression function's local behavior depends on angular relationships, such as shape analysis or directional statistics.

First, the following lemma for the angle comparison in $\text{CAT}(K)$ spaces is provided.

Lemma 6. Let $(\mathcal{M}, d)$ be a $\text{CAT}(K)$ space, and let $\triangle xyz \subset \mathcal{M}$ be a geodesic triangle of perimeter $< \pi/\sqrt{K}$ when $K > 0$. Let $\triangle \bar{x}\bar{y}\bar{z}$ be its comparison triangle in the simply connected model space of constant curvature K . Then for each vertex x and the corresponding comparison vertex $\bar{x}$, $\angle_x(y, z) \leq \angle_{\bar{x}}(\bar{y}, \bar{z})$, where $\angle_x(y, z)$ is the Alexandrov angle (or geodesic angle) at x formed by the geodesic segments $[xy]$ and $[xz]$.

Note the assumption that the perimeter of $\triangle xyz$ is $< \pi/\sqrt{K}$ (when $K > 0$) is used to ensure i) The geodesics $[xy], [yz], [zx]$ are short enough so that the entire triangle $\triangle xyz$ (and sub-triangles $\triangle xy'z'$) can be compared in the standard simply connected model space (the sphere of radius $1/\sqrt{K}$ if $K > 0$). ii) One avoids the potential degeneracy where side lengths might exceed $\pi/\sqrt{K}$, which could cause the model triangle in spherical geometry to become ambiguous or wrap around the sphere. In the case $K < 0$, there is no maximum perimeter restriction because the simply connected model space (Euclidean or hyperbolic) is unbounded in diameter.

Next, the lemma for the angle continuity under small perturbation is provided.

Lemma 7. Let $\triangle pqr$ and $\triangle p'q'r'$ be two geodesic triangles in a $\text{CAT}(K)$ space $(\mathcal{M}, d)$. Suppose each has a perimeter $\pi/\sqrt{K}$ when $K > 0$ (no restriction is needed if $K < 0$). Also assume $d(p, p') + d(q, q') + d(r, r')$ is small. Then, for the angles at p in $\triangle pqr$ and at p' in $\triangle p'q'r'$,

$$|\angle_p(q, r) - \angle_{p'}(q', r')| \leq C \delta_{pp'qq'rr'}. \quad (11)$$

where $C > 0$ is a constant depending only on K and the maximum side length (or perimeter) constraints, and

$$\delta_{pp'qq'rr'} := d(p, p') + d(q, q') + d(r, r'). \quad (12)$$

Based on the above lemmas, the following statements are obtained.

Proposition 3 (Angle perturbation via conditional measures). Let $\{\nu_x\}$ be a family of probability measures on a $\text{CAT}(K)$ space $(\mathcal{M}, d)$, each supported in a geodesic ball of diameter $< D = \pi/2\sqrt{K}$ when $K > 0$. Let $\mu^*(x)$ be the unique Fréchet mean of ν_x . Suppose ν_x and $\nu_{x'}$ are close in the Wasserstein metric on measures: $d_W(\nu_x, \nu_{x'}) \leq \epsilon$. Then, for any fixed $u, v \in \mathcal{M}$, one has

$$|\angle_{\mu^*(x)}(u, v) - \angle_{\mu^*(x')}(u, v)| \leq C\epsilon$$

where the constant $C > 0$ depends on the strong-convexity modulus $\alpha(K, D)$. In particular, smaller ϵ implies the angles at $\mu^*(x)$ and $\mu^*(x')$ to points u, v differ by at most $O(\epsilon)$.

Theorem 4 (Angle stability for conditional Fréchet means). Let $\{(X_i, Y_i)\} \subset \mathbb{R}^d \times \mathcal{M}$ with $\mathcal{M}$ a $\text{CAT}(K)$ space of diameter $< D = \pi/2\sqrt{K}$ if $K > 0$. For each $x \in \mathbb{R}^d$, let $\nu_x(\cdot)$ be the conditional distribution of Y given $X = x$. Assume each ν_x has the unique Fréchet mean $\mu^*(x)$. Moreover, suppose that for x, x' sufficiently close, the measures $\mu^*(x)$ and $\mu^*(x')$ differ by at most $\epsilon(\|x - x'\|)$ in the Wasserstein distance. Then for any finite set of points $\{u_1, \dots, u_m\} \subset \mathcal{M}$,

$$\sup_{1 \leq i < j \leq m} |\angle_{\mu^*(x)}(u_i, u_j) - \angle_{\mu^*(x')}(u_i, u_j)| \leq C\epsilon_{xx'}$$

where $C > 0$ is a constant depending on the strong-convexity modulus $\alpha(K, D)$ and $\epsilon_{xx'} := \epsilon(\|x - x'\|)$. Thus, all angles at $\mu^*(x)$ relative to a finite set of directions $u_1, \dots, u_m$ vary continuously and Lipschitzly with x .

Implications: The established angle stability results in Section 3.3 imply that the geometric structure surrounding the conditional Fréchet mean remains consistent under minor changes in the data distribution. This consistency is essential for applications where the relative orientation of data points carries meaningful information, ensuring that the regression estimates preserve intrinsic geometric relationships.

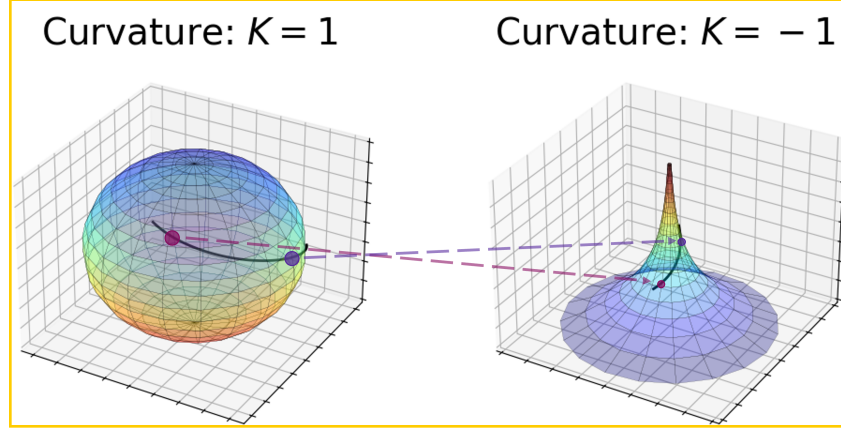


Figure 1: Mapping from spherical data into hyperbolic space.

3.4 Local Jet Expansion of Fréchet Functionals

Lemma 8. Let $z \in \mathcal{M}$ and let $\exp_z : T_z \mathcal{M} \rightarrow \mathcal{M}$ be the Riemannian exponential map (in a local sense if $\mathcal{M}$ is a manifold, or a suitable geodesic parameterization if $\mathcal{M}$ is just a geodesic metric space). Then for points u, v sufficiently close to z , define $U := \exp_z^{-1}(u)$ and $V := \exp_z^{-1}(v)$. Then,

$$\angle_z(u, v) = \angle_0(U, V) + O(\|\exp_z^{-1}(u)\|^2 + \|\exp_z^{-1}(v)\|^2),$$

where $\angle_0(U, V)$ is the standard Euclidean angle in $T_z \mathcal{M} \approx \mathbb{R}^m$, and the big-Oh term depends on curvature bounds near z .

Proposition 4 (Local Jet expansion of Fréchet functionals). Let ν be a probability measure on a sufficiently regular CAT(K) space $(\mathcal{M}, d)$. Suppose that $\mu(x)$ is the Fréchet mean of ν : $\mu(x) := \arg \min_{z \in \mathcal{M}} \int d^2(y, z) d\nu(y)$, and consider the Fréchet functional $F_x(z) = \int d^2(y, z) d\nu(y)$. Then, in a sufficiently small neighborhood of μ , the functional F can be expanded in the tangent space $T_\mu \mathcal{M}$ via the exponential map. Specifically, using local coordinates $\exp_\mu : T_\mu \mathcal{M} \supset B_r(0) \rightarrow \mathcal{M}$, for a vector v with $\|v\|$ small, define $z = \exp_\mu(v)$. The expansion is given by

$$F(\exp_\mu(v)) = F_x(\mu) + \langle \nabla F_x(\mu), v \rangle + \frac{1}{2} \langle H_x v, v \rangle + R(v),$$

where $\nabla F_x(\mu)$ is the gradient (which is zero if μ is the unique minimizer), H_x is the Hessian (a linear operator on $T_\mu \mathcal{M}$), and the remainder term $R(v)$ satisfies $|R(v)| = O(\|v\|^3)$.

Implications: The analysis in Section 3.4 offers a nuanced understanding of the Fréchet functional's local behavior around its minimizer, the Fréchet mean. By expanding the Fréchet functional in the tangent space via the exponential map, one can gain insights into the functional's curvature and higher-order properties.

3.5 Auxiliary Statements

Here, a couple of auxiliary propositions that facilitate a deeper understanding of the structural properties of the Fréchet functional within CAT(K) spaces are introduced in this section. These propositions decompose the Fréchet functional into radial and angular components, enabling a more nuanced analysis of variance and stability around the Fréchet mean.

Proposition 5 (Angle Splitting in Distance Sums). Consider the Fréchet functional $F(z) = \int d^2(y, z) d\nu(y)$. For z near μ^* , decompose:

$$d^2(y, z) = d^2(y, \mu^*) + \Pi_d(y, z, \mu^*) + \Pi_\angle(y, z, \mu^*),$$

where Π_d captures radial changes in distances, $\Pi_\angle$ represents angular corrections around μ^* . If $\angle_{\mu^*}(y, z)$ remains small near μ^* , then $\Pi_\angle$ is of order $\langle \angle_{\mu^*}(y, z) \rangle d(\mu^*, z)$.

Proposition 6 (Angle-Distance Decomposition of Conditional Variance). Let ν_x be the conditional distribution of Y given $X = x$ on a sufficiently smooth CAT(K) space $(\mathcal{M}, d)$. Suppose $\mu^*(x)$ is

Data manifold	Mean squared error (MSE)
Sphere ($K = 1$)	0.4915(± 0.0086)
Hyperbolic ($K = -1$)	0.4228(± 0.0021)

Table 1: Evaluation of Fréchet regression on different spaces.

the unique Fréchet mean of ν_x . Around $\mu^*(x)$, let

$$R_x(y) := d(y, \mu^*(x)), \quad \phi_x(y) := \angle_{\mu^*(x)}(u_0, y), \quad (13)$$

for a fixed reference point $u_0 \in \mathcal{M}$. Then the conditional variance can be partially decomposed into a radial variance term, an angle-radial covariance term, and higher-order corrections:

$$\begin{aligned} \text{Var}_{\nu_x} [d^2(Y, \mu^*(x))] \\ = \text{Var}[A_x(Y)] + \text{Cov}(\phi_x(Y), R_x(Y)^2) + \beta, \end{aligned} \quad (14)$$

where A_x is the radial part and β is the higher-order term.

Implications: The auxiliary propositions presented in Subsection 3.5 play an important role in refining the theoretical underpinnings of Fréchet regression within $\text{CAT}(K)$ spaces. By decomposing the Fréchet functional into radial and angular components, these propositions enable a more granular analysis of variance and stability around the Fréchet mean.

4 Experiments

From the discussion in Section 3, it can be seen that the negative curvature space has better properties in terms of estimation than the positive curvature space with broader support. To confirm these results, this section considers numerical experiments. See Appendix A for the intuitive understanding of the following hyperbolic mapping.

4.1 Illustrative Example

A point on the unit sphere is parameterized as $x = \sin(\phi) \cos(\theta)$, $y = \sin(\phi) \sin(\theta)$, $z = \cos(\phi)$, where $\phi \in [0, \pi]$ is the polar angle and $\theta \in [0, 2\pi]$ is the azimuthal angle. Let R be the radius of the sphere. Here, consider the stereographic projection: The plane is tangent to the sphere at the south pole $(0, 0, -R)$ and is defined $z = -R$, and the north pole $N = (0, 0, R)$ serves as the projection point. For a point $p = (x, y, z)$, the stereographic projection $\pi(p) = (u, v)$ on the plane is given by $u = \frac{Rx}{R+z}$, $v = \frac{Ry}{R+z}$. This plane can be considered in the hyperbolic space, and one can visualize it as the pseudosphere (see Figure 1). Also, a point (x, y, z) can be mapped back to the sphere as

$$x = \frac{2R^2u}{R^2 + u^2 + v^2}, y = \frac{2R^2v}{R^2 + u^2 + v^2}, z = R \frac{u^2 + v^2 - R^2}{R^2 + u^2 + v^2}.$$

See Appendix E (including Python code in Listing 2) for the detailed data-generating process.

Table 1 shows the evaluation results of Fréchet regression on the spherical and hyperbolic coordinates. It can be seen that the hyperbolic mapping yields better results. Note that, the previous studies [15, 16] reported the effectiveness of such mapping for statistical problems of spherical data, and the objective of experiments in this section is just to confirm the theoretical results.

4.2 Experiment on Real-world Dataset

In addition to the illustrative example, consider the experiments on the real-world datasets. This section uses the following: i) HYG Steller database¹, which is a comprehensive dataset containing information on stars brighter than magnitude 6.5. ii) USGS Earthquake catalogue², represented in spherical coordinates. iii) NOAA Climate data³, from weather satellites. See Appendix 4.2 for the

¹<https://github.com/astronexus/HYG-Database?tab=readme-ov-file>

²https://earthquake.usgs.gov/earthquakes/feed/v1.0/summary/2.5_week.csv

³<http://celestrak.org/NORAD/elements/table.php?GROUP=weather&FORMAT=t1e>

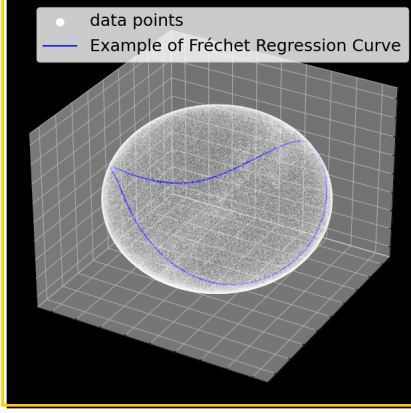


Figure 2: Visualization of the HYG Stellar database.

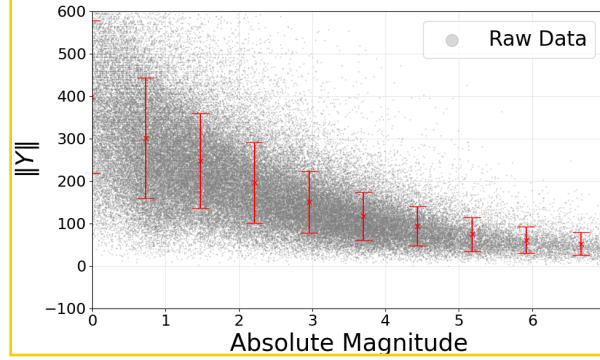


Figure 3: Heteroscedasticity in the HYG Stellar dataset.

Dataset	MSE
HYG Stellar	0.3765(± 0.0036)
USGS Earthquake	0.5832(± 0.0831)
NOAA Climate	0.4384(± 0.0678)
HYG Stellar (hyperbolic)	0.2660(± 0.0032)
USGS Earthquake (hyperbolic)	0.4743(± 0.0541)
NOAA Climate (hyperbolic)	0.3259(± 0.0683)

Table 2: Evaluation of Fréchet regression on different spaces.

details of this experiment (including Python code in Listing 3 for the visualization and data format check of the dataset). Table 2 shows the experimental results of Fréchet regression on different coordinates for the real datasets. The mapping procedure is the same as Section 4.1. As with the illustrative example, we can confirm that Fréchet regression on hyperbolic surfaces yields better results on the real datasets. As discussed in more detail in Appendix A, such a mapping of responses to hyperbolic space may be particularly useful when heteroscedasticity is assumed in the data. Indeed, heteroscedasticity can be observed in the HYG Stellar dataset (see Figure 3).

5 Conclusion

This study provides a comprehensive theoretical analysis of Fréchet regression within the framework of comparison geometry, focusing on $\text{CAT}(K)$ spaces. It establishes foundational results on the existence, uniqueness, and stability of the Fréchet mean under varying curvature conditions. Notably, the analysis demonstrates how curvature properties influence statistical estimation, with non-positive curvature spaces offering advantageous stability and convergence properties. The paper also extends statistical guarantees to nonparametric Fréchet regression, including exponential concentration bounds and convergence rates, which align with classical Euclidean results. Angle stability and local jet expansion further highlight the behavior of Fréchet functionals, offering geometric insights of regression in non-Euclidean spaces. Experimental results support the theoretical findings, showing that hyperbolic mappings often improve performance under heteroscedasticity assumption.

Limitations: While this study provides a robust theoretical foundation for Fréchet regression in $\text{CAT}(K)$ spaces, several limitations exist. Firstly, the analysis predominantly focuses on spaces with constant curvature bounds, which may not encompass all practical scenarios where data resides in more heterogeneous geometric contexts. Additionally, the reliance on strong convexity conditions and diameter constraints in positively curved spaces may restrict the applicability of the results. As has been done in the information geometry framework [1, 34, 10, 25, 26, 31, 2], future work could explore relaxing assumptions, extending the framework to broader classes of metric spaces, and developing efficient algorithms.

References

- [1] Shotaro Akaho. The e-pca and m-pca: Dimension reduction of parameters by information geometry. In *2004 IEEE International Joint Conference on Neural Networks (IEEE Cat. No. 04CH37541)*, volume 1, pages 129–134. IEEE, 2004.
- [2] Shun-Ichi Amari. Natural gradient works efficiently in learning. *Neural computation*, 10(2): 251–276, 1998.
- [3] Shun-ichi Amari. *Information geometry and its applications*, volume 194. Springer, 2016.
- [4] Shun-ichi Amari and Hiroshi Nagaoka. *Methods of information geometry*, volume 191. American Mathematical Soc., 2000.
- [5] Nihat Ay, Jürgen Jost, Hồng Vân Lê, and Lorenz Schwachhöfer. *Information geometry*, volume 64. Springer, 2017.
- [6] Werner Ballmann. *Lectures on spaces of nonpositive curvature*, volume 25. Springer Science & Business Media, 1995.
- [7] Satarupa Bhattacharjee and Hans-Georg Müller. Single index fréchet regression. *The Annals of Statistics*, 51(4):1770–1798, 2023.
- [8] Hermanus Josephus Bierens. The nadaraya-watson kernel regression function estimator. 1988.
- [9] Martin R Bridson and André Haeffliger. *Metric spaces of non-positive curvature*, volume 319. Springer Science & Business Media, 2013.
- [10] Kevin M Carter, Raviv Raich, William G Finn, and Alfred O Hero III. Information-geometric dimensionality reduction. *IEEE Signal Processing Magazine*, 28(2):89–99, 2011.
- [11] Jeff Cheeger and Karsten Grove. *Metric and comparison geometry*, volume 11. International Press, 2007.
- [12] Jeff Cheeger, David G Ebin, and David Gregory Ebin. *Comparison theorems in Riemannian geometry*, volume 9. North-Holland publishing company Amsterdam, 1975.
- [13] Yaqing Chen and Hans-Georg Müller. Uniform convergence of local fréchet regression with applications to locating extrema and time warping for metric space valued trajectories. *The Annals of Statistics*, 50(3):1573–1592, 2022.
- [14] Brad C Davis, P Thomas Fletcher, Elizabeth Bullitt, and Sarang Joshi. Population shape regression from random design data. *International journal of computer vision*, 90:255–266, 2010.
- [15] TD Downs. Spherical regression. *Biometrika*, 90(3):655–668, 2003.
- [16] Kajal Eybpoosh, Mansoor Rezaghi, and Abbas Heydari. Applying inverse stereographic projection to manifold learning and clustering. *Applied Intelligence*, pages 1–15, 2022.
- [17] Daniel Ferguson and François G Meyer. Computation of the sample fréchet mean for sets of large graphs with applications to regression. In *Proceedings of the 2022 SIAM International Conference on Data Mining (SDM)*, pages 379–387. SIAM, 2022.
- [18] Aritra Ghosal. *Application of the single index methodology to the local Fréchet regression in the context of Object oriented data analysis (OODA)*. University of California, Santa Barbara, 2023.
- [19] Aritra Ghosal, Wendy Meiring, and Alexander Petersen. Fréchet single index models for object response regression. *Electronic Journal of Statistics*, 17(1):1074–1112, 2023.
- [20] Karsten Grove and Peter Petersen. *Comparison geometry*, volume 30. Cambridge University Press, 1997.
- [21] Matthias Hein. Robust nonparametric regression with metric-space valued output. *Advances in neural information processing systems*, 22, 2009.

- [22] Jürgen Jost. *Nonpositive curvature: geometric and analytic aspects*. Birkhäuser, 2012.
- [23] Hermann Karcher. Riemannian center of mass and mollifier smoothing. *Communications on pure and applied mathematics*, 30(5):509–541, 1977.
- [24] David G Kendall. Shape manifolds, procrustean metrics, and complex projective spaces. *Bulletin of the London mathematical society*, 16(2):81–121, 1984.
- [25] Masanari Kimura. Generalized t-sne through the lens of information geometry. *IEEE Access*, 9: 129619–129625, 2021.
- [26] Masanari Kimura and Howard Bondell. Density ratio estimation via sampling along generalized geodesics on statistical manifolds. *arXiv preprint arXiv:2406.18806*, 2024.
- [27] Masanari Kimura and Hideitsu Hino. α -geodesical skew divergence. *Entropy*, 23(5):528, 2021.
- [28] Masanari Kimura and Hideitsu Hino. Information geometrically generalized covariate shift adaptation. *Neural Computation*, 34(9):1944–1977, 2022.
- [29] Zhenhua Lin and Hans-Georg Müller. Total variation regularized fréchet regression for metric-space valued data. *The Annals of Statistics*, 49(6):3510–3533, 2021.
- [30] Dong C Liu and Jorge Nocedal. On the limited memory bfgs method for large scale optimization. *Mathematical programming*, 45(1):503–528, 1989.
- [31] Noboru Murata, Takashi Takenouchi, Takafumi Kanamori, and Shinto Eguchi. Information geometry of u-boost and bregman divergence. *Neural Computation*, 16(7):1437–1481, 2004.
- [32] Elizbar A Nadaraya. On estimating regression. *Theory of Probability & Its Applications*, 9(1): 141–142, 1964.
- [33] Frank Nielsen. An elementary introduction to information geometry. *Entropy*, 22(10):1100, 2020.
- [34] Adrian M Peter and Anand Rangarajan. Information geometry for landmark shape analysis: Unifying shape representation and deformation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 31(2):337–350, 2008.
- [35] Alexander Petersen and Hans-Georg Müller. Fréchet regression for random objects with euclidean predictors. *The Annals of Statistics*, 47(2):691–719, 2019.
- [36] Rui Qiu, Zhou Yu, and Ruqing Zhu. Random forest weighted local fréchet regression with random objects. *Journal of Machine Learning Research*, 25(107):1–69, 2024.
- [37] Dogyoon Song and Kyunghye Han. Errors-in-variables fréchet regression with low-rank covariate approximation. *Advances in Neural Information Processing Systems*, 36:80575–80607, 2023.
- [38] Florian Steinke and Matthias Hein. Non-parametric regression between manifolds. *Advances in neural information processing systems*, 21, 2008.
- [39] Danielle C Tucker, Yichao Wu, and Hans-Georg Müller. Variable selection for global fréchet regression. *Journal of the American Statistical Association*, 118(542):1023–1037, 2023.
- [40] Geoffrey S Watson. Smooth regression analysis. *Sankhyā: The Indian Journal of Statistics, Series A*, pages 359–372, 1964.
- [41] Guofang Wei and Will Wylie. Comparison geometry for the bakry-emery ricci tensor. *Journal of differential geometry*, 83(2):337–405, 2009.
- [42] Xingyu Yan, Xinyu Zhang, and Peng Zhao. Frequentist model averaging for global fréchet regression. *IEEE Transactions on Information Theory*, 2024.
- [43] Takumi Yokota. Convex functions and barycenter on $\text{cat}(1)$ -spaces of small radii. *Journal of the Mathematical Society of Japan*, 68(3):1297–1323, 2016.
- [44] Qi Zhang, Lingzhou Xue, and Bing Li. Dimension reduction for fréchet regression. *Journal of the American Statistical Association*, 119(548):2733–2747, 2024.

A Intuitive Understanding for Hyperbolic Mapping

In regression analysis, transforming the response variable can often lead to improved model performance by stabilizing variance, normalizing distributions, or linearizing relationships. A classical example is the logarithmic transformation $Y \mapsto \log(Y)$ which can enhance the performance of a linear regression model under certain conditions. Similarly, mapping spherical responses into hyperbolic space can offer analogous benefits, particularly in scenarios where the data exhibits inherent geometric or hierarchical structures.

Log Transformation in Linear Regression Consider the simple linear regression model:

$$Y = \beta X + \epsilon,$$

where Y is the response variable, X is the predictor, β is the regression coefficient, and ϵ is the error term with $\mathbb{E}[\epsilon] = 0$ and $\text{Var}(\epsilon) = \sigma^2$. Applying a logarithmic transformation to Y yields

$$\begin{aligned} \log(Y) &= \beta X + \epsilon, \\ Y &= \exp(\beta X + \epsilon) = \exp(\beta X) \cdot \exp(\epsilon). \end{aligned}$$

Assuming ϵ is small and approximately normally distributed, $\exp(\epsilon)$ introduces multiplicative noise to Y effectively stabilizing variance across different levels of X . This transformation often reduces heteroscedasticity in the residuals, leading to improved regression performance. Here, the heteroscedasticity refers to the phenomenon where the variability of the errors (or residuals) in a regression model is not constant across the range of predictor variables.

Definition 8 (Heteroscedasticity). Consider a regression model:

$$Y_i = \beta X_i + \epsilon_i,$$

where $\epsilon_i \sim \mathcal{N}(0, \sigma^2(X_i))$. Here, the variance of the error term $\sigma^2(X)$ depends on X . In a heteroscedastic model, the variance of ϵ_i is a function of the predictors X_i :

$$\text{Var}(\epsilon_i | X_i) = \sigma^2(X_i).$$

In contrast, for homoscedasticity, the variance of ϵ_i is constant.

Hyperbolic Mapping via Stereographic Projection Analogous to the log transformation, hyperbolic mapping transforms the response variable into a space where the geometric structure can lead to improved regression characteristics. The procedure involves mapping points from a spherical representation to a hyperbolic plane using stereographic projection. A point on the unit sphere of radius R is parameterized using spherical coordinates:

$$\begin{aligned} x &= R \sin(\phi) \cos(\theta), \\ y &= R \sin(\phi) \sin(\theta), \\ z &= R \cos(\phi), \end{aligned}$$

where $\phi \in [0, \pi]$ is the polar angle and $\theta \in [0, 2\pi]$ is the azimuthal angle. The stereographic projection maps a point $p = (x, y, z)$ on the sphere to a point $p \mapsto \psi(p) = (u, v)$ on the plane tangent to the sphere at the south pole $(0, 0, -R)$ and defined by $z = -R$. The north pole $N = (0, 0, R)$ serves as the projection point. The projection formulas are

$$\begin{aligned} u &= \frac{Rx}{R + z}, \\ v &= \frac{Ry}{R + z}. \end{aligned}$$

This plane can be interpreted as a model of hyperbolic space, specifically visualized as a pseudosphere, which inherently possesses properties conducive to handling hierarchical or tree-like data structures.

Both the logarithmic transformation and hyperbolic mapping aim to stabilize variance and linearize relationships, through different geometric transformations. To understand the benefits of hyperbolic mapping, consider the effect of each transformation on the variance of the response variable. Starting with $Y = \beta X + \epsilon$, applying the log transformation yields

$$\log Y = \beta X + \epsilon.$$

451 Assuming $\epsilon \sim \mathcal{N}(0, \sigma^2)$. The variance of $\log Y$ remains σ^2 which can be advantageous if the original
 452 Y exhibits multiplicative noise:

$$\text{Var}(Y) = \text{Var}(\exp(\beta X + \epsilon)) = \exp(2\beta X) \cdot (\exp(\sigma^2) - 1).$$

453 The transformation effectively decouples the variance from X stabilizing it across different predictor
 454 values.

455 For hyperbolic mapping, consider a response variable represented as a point on the sphere. The
 456 stereographic projection transforms this spherical representation into the hyperbolic plane. Let
 457 Y be the original response mapped to a point $p = (x, y, z)$ on the sphere, and $\psi(p) = (u, v)$ its
 458 hyperbolic projection. Assuming small deviations around a mean direction, the hyperbolic mapping
 459 can linearize angular variations similarly to how the log transformation linearizes multiplicative
 460 variations. Specifically, fluctuations in Y around the mean direction correspond to additive noise
 461 in the hyperbolic plane, potentially reducing variance in a manner akin to the log transformation.
 462 Formally, if Y is modeled on the sphere with

$$Y = R \cdot p + \epsilon,$$

463 where ϵ represents angular noise, the hyperbolic projection yields

$$\psi(Y) = \left(\frac{Rx}{R+z}, \frac{Ry}{R+z} \right) + \epsilon',$$

464 where ϵ' is the transformed noise. Under specific conditions (e.g., small angular deviations), ϵ'
 465 exhibits reduced variance compared to ϵ , analogous to the variance stabilization achieved by the log
 466 transformation.

467 **Example 1** (Stabilizing Variance in Hierarchical Data). Consider a dataset where the response
 468 variable Y represents hierarchical relationships, such as the popularity of topics in a taxonomy. The
 469 inherent tree-like structure implies that differences between nodes (topics) grow exponentially with
 470 depth. Direct regression on Y would face increasing variance as depth increases. By mapping Y into
 471 hyperbolic space via stereographic projection, the exponential growth inherent in hierarchical data
 472 is linearized. This transformation stabilizes variance across different levels of the hierarchy, enabling
 473 more effective regression modeling. Specifically, the hyperbolic mapping aligns the geometric
 474 properties of the data with the regression framework, similar to how the log transformation aligns
 475 multiplicative relationships with additive modeling.

476 Let Y be mapped to hyperbolic space via stereographic projection:

$$u = \frac{Rx}{R+z},$$

$$v = \frac{Ry}{R+z}.$$

477 Assuming Y lies close to the north pole $N = (0, 0, R)$, small perturbations ϵ around N imply

$$z = R \cos(\phi) \approx R \left(1 - \frac{\phi^2}{2} \right),$$

$$x = R \sin(\phi) \cos(\theta) \approx R\phi \cos(\theta),$$

$$y = R \sin(\phi) \sin(\theta) \approx R\phi \sin(\theta).$$

478 Substituting into the projection formulas,

$$u \approx \frac{R \cdot R\phi \cos(\theta)}{R + R \left(1 - \frac{\phi^2}{2} \right)} = \frac{R^2 \phi \cos(\theta)}{2R - \frac{\phi^2}{2}} \approx \frac{R\phi \cos(\theta)}{2},$$

$$v \approx \frac{R \cdot R\phi \sin(\theta)}{R + R \left(1 - \frac{\phi^2}{2} \right)} = \frac{R^2 \phi \sin(\theta)}{2R - \frac{\phi^2}{2}} \approx \frac{R\phi \sin(\theta)}{2}.$$

479 Thus, small angular deviations ϕ result in approximately linear changes in u and v , effectively
 480 reducing the variance from multiplicative to additive in the hyperbolic plane:

$$\text{Var}(u, v) \approx \left(\frac{R}{2} \right)^2 \text{Var}(\phi).$$

481 Compared to the original spherical variance $\text{Var}(\phi)$, the hyperbolic mapping scales and linearizes the
 482 variance, analogous to the stabilizing effect of the log transformation. Figure 4 shows the illustrative
 483 example of transformed responses for $Y = \beta X + \epsilon$ with heteroscedastic errors $\epsilon = \mathcal{N}(0, g(\sigma X))$,
 484 $\sigma = 0.2$ and $\beta = 2$. This figure shows $g(\sigma X) = \sigma X$ and $g(\sigma X) = \exp(\sigma X)$ cases.

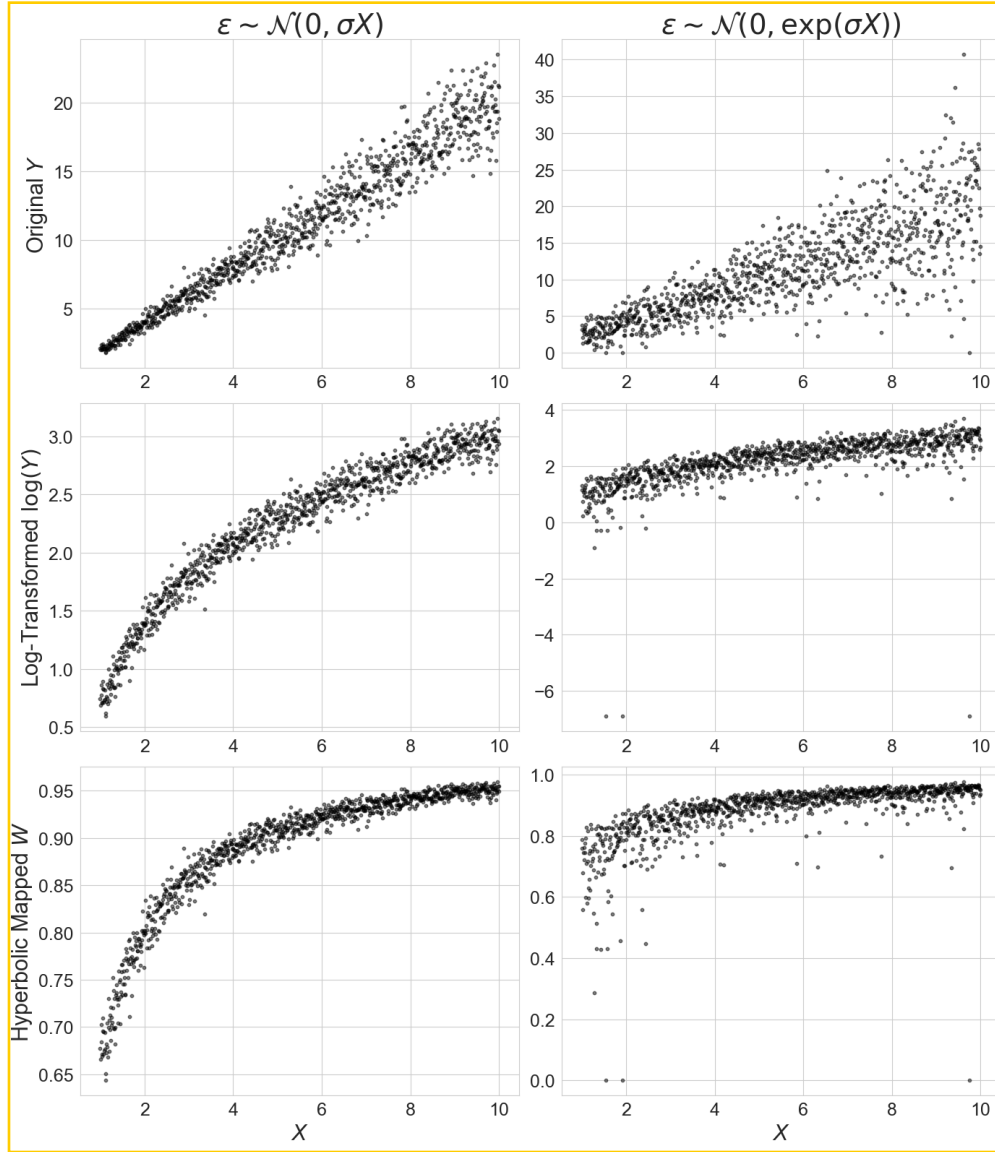


Figure 4: Illustrative example of transformed responses. Under the heteroscedastic errors assumption, the appropriate transformations of response variable yield stabilized variance. In this figure, Y is the original response variables, $\log(Y)$ is the log-transformed variables and W is the hyperbolic mapped variables.

485 B Proofs

486 B.1 Proofs for Section 3.1

487 *Proof for Lemma 1.* To establish the geodesic convexity of the squared distance function $f(x) =$
 488 $d^2(p, x)$ in a $\text{CAT}(K)$ space $(\mathcal{M}, d)$ with $K < 0$, one must show that for any two points $x, y \in \mathcal{M}$
 489 and any geodesic $\gamma: [0, 1] \rightarrow \mathcal{M}$ connecting x to y , the function $t \mapsto f(\gamma(t))$ is convex on the
 490 interval $[0, 1]$.

491 In the model space $\mathbb{M}_K^2$ of constant curvature $K < 0$, construct a comparison triangle $\bar{\Delta}$ correspond-
 492 ing to $\Delta = \{p, x, y\}$ in $\mathcal{M}$. Let $\bar{p}, \bar{x}, \bar{y}$ be the vertices of $\bar{\Delta}$ in $\mathbb{M}_K^2$ with side lengths matching those
 493 of Δ . Then, for any points a, b on the sides $[x, y]$ and $[p, x]$ or $[p, y]$, the distance $d(a, b)$ in $\mathcal{M}$ is at
 494 most the distance $d_{\mathbb{M}_K^2}(a, b)$ in the model space.

495 Let $\gamma(t)$ corresponds to a point $\bar{\gamma}(t)$ on the side $[\bar{x}, \bar{y}]$ in $\bar{\Delta}$. By the $\text{CAT}(K)$ property,

$$d(p, \gamma(t)) \leq d_{\mathbb{M}_K^2}(\bar{p}, \bar{\gamma}(t)).$$

496 In $\mathbb{M}_K^2$, which is a uniquely geodesic space, the squared distance satisfies the law of cosines

$$d^2(\bar{p}, \bar{\gamma}(t)) \leq (1-t)d^2(\bar{p}, \bar{x}) + td^2(\bar{p}, \bar{y}) - t(1-t)c_K$$

497 where c_K is a non-negative constant dependent on K and the geometry of the triangle. Here, since
 498 $K < 0$, the space $\mathbb{M}_K^2$ exhibits non-positive curvature, which implies that the term $-t(1-t)c_K$ does
 499 not negatively affect the inequality. Therefore,

$$d^2(p, \gamma(t)) \leq d_{\mathbb{M}_K^2}^2(\bar{p}, \bar{\gamma}(t)) \leq (1-t)d^2(p, x) + td^2(p, y).$$

500 and f is geodesically convex. □

501 *Proof for Lemma 2.* Consider a sequence $\{x_n\}$ in $\mathcal{M}$ that converges to $x \in \mathcal{M}$. Given the continuity
 502 of the distance function in metric spaces, for each $y \in \mathcal{M}$, $d(y, x_n) \rightarrow d(y, x)$ as $n \rightarrow +\infty$. Since
 503 $d^2(y, x)$ is continuous in x , by Fatou's lemma,

$$\liminf_{n \rightarrow +\infty} d^2(y, x_n) \leq d^2(y, x).$$

504 Integrating both sides with respect to ν ,

$$\liminf_{n \rightarrow +\infty} \int_{\mathcal{M}} d^2(y, x_n) d\nu(y) \leq \int_{\mathcal{M}} d^2(y, x) d\nu(y).$$

505 Thus, F is lower semicontinuous. Also, since

$$F(x) = \int_{\mathcal{M}} d^2(y, x) d\nu(y) \geq 0,$$

506 for any $x \in \mathcal{M}$, F is bounded below by zero. Therefore, there exists a sequence $\{m_n\}$ in $\mathcal{M}$ such
 507 that

$$F(m_n) \rightarrow \inf_{x \in \mathcal{M}} F(x),$$

508 as $n \rightarrow +\infty$. Let $\{m_n\}$ be called a minimizing sequence. Given that the support of ν , denoted by
 509 $\text{supp}(\nu)$, is compact, denote it by $S \subseteq \mathcal{M}$. That is, S is compact and $\nu(S) = 1$.

510 To ensure that the existence of a convergent subsequence, one need to show that $\{m_n\}$ is contained
 511 within a compact subset of $\mathcal{M}$. Since S is compact, it is bounded. Thus, there exists a radius $R > 0$
 512 and a point $p \in \mathcal{M}$ such that $S \subset B(p, R)$, where $B(p, R) = \{x \in \mathcal{M} \mid d(p, x) \leq R\}$. Using the
 513 triangle inequality in metric spaces,

$$d(y, m_n) \geq d(p, m_n) - d(y, p) \geq d(p, m_n) - R.$$

514 Then,

$$\begin{aligned} F(m_n) &= \int_S d^2(y, m_n) d\nu(y) \\ &\geq \int_S \{d(p, m_n) - d(y, p)\}^2 d\nu(y) \\ &= \int_S \{d(p, m_n)^2 - 2d(p, m_n)d(y, p) + d^2(y, p)\} d\nu(y) \\ &= d(p, m_n)^2 - 2d(p, m_n) \int_S d(y, p) d\nu(y) + \int_S d^2(y, p) d\nu(y) \leq C \end{aligned}$$

515 Let $A = \int_S d(y, p) \nu(y)$ and $B = \int_S d^2(y, p) \nu(y)$, both finite due to the compactness. Thus,

$$\begin{aligned} d(p, m_n)^2 - 2Ad(p, m_n) + B &\leq C \\ d(p, m_n) &\leq A \pm \sqrt{A^2 + C - B}. \end{aligned}$$

516 Hence, the sequence $\{m_n\}$ lies within the closed ball $\bar{B}(p, A + \sqrt{A^2 + C - B})$, which is compact
 517 if $\mathcal{M}$ is proper. Here, $\text{CAT}(K)$ spaces are not necessarily proper in general, but since $\text{supp}(\nu)$ is
 518 compact and $\{m_n\}$ is bounded, one can extract a convergent subsequence under the assumption
 519 that $\mathcal{M}$ is complete. Given that $\{m_n\}$ is bounded and $\mathcal{M}$ is complete, one can utilize the Bolzano-
 520 Weierstrass theorem in $\text{CAT}(K)$ spaces to extract a convergent subsequence. Specifically, since $\mathcal{M}$
 521 is a geodesic space and $\{m_n\}$ is bounded, there exists a subsequence $\{m_{n_k}\}$ that converges to some
 522 $m \in \mathcal{M}$.

523 Since F is lower semicontinuous and $m_{n_k} \rightarrow m$,

$$F(m) \leq \liminf_{k \rightarrow \infty} F(m_{n_k}) = \inf_{x \in \mathcal{M}} F(x).$$

524 This implies that m achieves the infimum of F .

$$F(m) = \inf_{x \in \mathcal{M}} F(x).$$

525 Therefore, m is a minimizer of the Fréchet functional. □

526 *Proof for Lemma 3.* For the sake of contradiction, suppose that there are two distinct points $m_1, m_2 \in$
 527 $\mathcal{M}$ such that both are minimizers of the Fréchet functional.

$$\begin{aligned} m_1 &= \arg \min_{x \in \mathcal{M}} \int_{\mathcal{M}} d^2(y, x) \nu(y), \\ m_2 &= \arg \min_{x \in \mathcal{M}} \int_{\mathcal{M}} d^2(y, x) \nu(y), \end{aligned}$$

528 with $m_1 \neq m_2$. Since $\mathcal{M}$ is a $\text{CAT}(K)$ space and thus a geodesic metric space, there exists a unique
 529 geodesic $\gamma: [0, 1] \rightarrow \mathcal{M}$ connecting m_1 to m_2 .

$$\begin{aligned} \gamma(0) &= m_1, \\ \gamma(1) &= m_2, \\ d(\gamma(t), \gamma(t')) &= |t - t'| \cdot d(m_1, m_2), \quad \forall t, t' \in [0, 1]. \end{aligned}$$

530 Define a function $F: [0, 1] \rightarrow \mathbb{R}$ by evaluating the Fréchet functional along the geodesic $\gamma(t)$:

$$F(t) = \int_{\mathcal{M}} d^2(y, \gamma(t)) \nu(y).$$

531 Since both m_1 and m_2 are minimizers,

$$F(0) = F(1) = \inf_{x \in \mathcal{M}} F(x).$$

532 Given that $\mathcal{M}$ is strictly geodesically convex, the squared distance function $f(x) = d^2(y, x)$ is strictly
 533 convex along any geodesic. Therefore, for each fixed $y \in \mathcal{M}$, the function $t \mapsto d^2(y, \gamma(t))$ satisfies

$$d^2(y, \gamma(t)) < (1 - t)d^2(y, m_1) + td^2(y, m_2),$$

534 for all $t \in (0, 1)$.

535 Integrate the strict inequality with respect to the measure ν yields

$$\begin{aligned} F(t) &= \int_{\mathcal{M}} d^2(y, \gamma(t)) \nu(y) \\ &< \int_{\mathcal{M}} \{(1 - t)d^2(y, m_1) + td^2(y, m_2)\} \nu(y) \\ &= (1 - t) \int_{\mathcal{M}} d^2(y, m_1) \nu(y) + t \int_{\mathcal{M}} d^2(y, m_2) \nu(y). \end{aligned}$$

536 But since m_1 and m_2 are both minimizers,

$$\int_{\mathcal{M}} d^2(y, m_1) d\nu(y) = \int_{\mathcal{M}} d^2(y, m_2) d\nu(y) = \int_{x \in \mathcal{M}} F(x).$$

537 Thus,

$$F(t) < (1-t) \inf_{x \in \mathcal{M}} F(x) + t \inf_{x \in \mathcal{M}} F(x) = \inf_{x \in \mathcal{M}} F(x).$$

538 However, this is a contradiction because $F(x)$ cannot be less than the infimum $\inf_{x \in \mathcal{M}} F(x)$. The
 539 contradiction arises from the assumption that two distinct minimizers m_1 and m_2 exist. Therefore,
 540 there can be at most one minimizer. Given that the Fréchet functional attains its infimum by Lemma 2,
 541 this minimizer is unique. $\square$

542 *Proof for Proposition 1.* The Fréchet functional $x \mapsto F_\nu(x)$ for a measure ν is defined as

$$F_\nu(x) = \int_{\mathcal{M}} d^2(y, x) d\nu(y).$$

543 Given that the squared distance function $d^2(y, x)$ is continuous in y for each fixed x , weak conver-
 544 gence $\nu_n \Rightarrow \nu$ implies that for each fixed $x \in \mathcal{M}$,

$$\lim_{n \rightarrow +\infty} F_{\nu_n}(x) = F_\nu(x).$$

545 In addition, given that $d^2(y, x)$ is continuous and bounded by zero, and assuming that the measures
 546 ν_n and ν have compact supports, as established in Lemma 2, the convergence $\nu_n \Rightarrow \nu$ implies that

$$\lim_{n \rightarrow +\infty} F_{\nu_n}(x) = F_\nu(x), \quad \text{uniformly for } x \in \mathcal{M}.$$

547 This uniform convergence is a consequence of the boundedness of the squared distance function
 548 over compact supports, and the equicontinuity provided by the geometric properties of the $\text{CAT}(K)$
 549 spaces.

550 Suppose that m_n does not converge to m . Then, there exist an $\epsilon > 0$ and a subsequence $\{m_{n_k}\}$ such
 551 that

$$d(m_{n_k}, m) \geq \epsilon,$$

552 for all k . Since $\mathcal{M}$ is a $\text{CAT}(K)$ space with $K \leq 0$ and hence a geodesic and proper metric space
 553 under the assumption of compact support from Lemma 2, the sequence $\{m_{n_k}\}$ has a convergent
 554 subsequence. Without loss of generality, assume that $m_{n_k} \rightarrow m'$ as $k \rightarrow +\infty$. By the continuity of
 555 the Fréchet functional,

$$\begin{aligned} \lim_{k \rightarrow +\infty} F_{\nu_{n_k}}(m_{n_k}) &= \lim_{k \rightarrow +\infty} \inf_{x \in \mathcal{M}} F_{\nu_{n_k}}(x) \\ &= F_\nu(m), \end{aligned}$$

556 since m is the unique minimizer for ν .

557 Consider $\nu_n \Rightarrow \nu$ and $m_{n_k} \rightarrow m'$.

$$\lim_{k \rightarrow +\infty} F_{\nu_{n_k}}(m_{n_k}) = F_\nu(m').$$

558 Then,

$$F_\nu(m') = F_\nu(m).$$

559 Therefore, m' is also a minimizer of $F_\nu(x)$. Since ν has a unique Fréchet mean m , it must be that
 560 $m' = m$. Recall that $d(m_{n_k}, m) \geq \epsilon$ for all k , but $m_{n_k} \rightarrow m' = m$, which implies that

$$\lim_{k \rightarrow +\infty} d(m_{n_k}, m) = d(m', m) = 0,$$

561 contradicting $d(m_{n_k}, m) \geq \epsilon$. Therefore, it must be that

$$m_n \rightarrow m, \quad \text{as } n \rightarrow +\infty.$$

562

$\square$

563 *Proof for Proposition 4.* For $K > 0$, the comparison space is the standard sphere $\mathbb{S}^n$ with radius
 564 $1/\sqrt{K}$. In $\mathbb{S}^n$, geodesics are great circles, and the distance between two points is given by the
 565 central angle multiplied by $1/\sqrt{K}$. The diameter of $\mathbb{S}^n$ is $\pi/\sqrt{K}$, meaning that the maximal distance
 566 between any two points is $\pi/\sqrt{K}$.

567 Given $R < \pi/2\sqrt{K}$, the geodesic ball $B(p, R)$ lies entirely within a hemisphere of $\mathbb{S}^n$. In this
 568 setting, any two points $x, y \in B(p, R)$ are separated by a distance $d(x, y)$, satisfying

$$\begin{aligned} d(x, y) &\leq d(x, p) + d(p, y) \\ &< \frac{\pi}{2\sqrt{K}} + \frac{\pi}{2\sqrt{K}} \\ &= \frac{\pi}{\sqrt{K}}. \end{aligned}$$

569 Since $d(x, y) < \pi/\sqrt{K}$, there exists a unique minimal geodesic connecting x and y within $\mathbb{S}^n$.

570 Assume, for contradiction, that the minimal geodesic γ between x and y exits $B(p, R)$. Then, there
 571 exists a point $z \in \gamma$ such that $d(p, z) = R$. Consider the geodesic triangles $\triangle pzx$ and $\triangle pzy$. Since
 572 $d(p, x) < R$ and $d(p, y) < R$, and γ is minimal, the angle at p opposite the side γ must satisfy certain
 573 angular constraints derived from the spherical law of cosines. However, because $R < \pi/2\sqrt{K}$, the
 574 triangle $\triangle pzx$ lies within a convex hemisphere, ensuring that the path from p to z to x remains within
 575 $B(p, R)$. This contradicts the assumption that γ exits $B(p, R)$. Therefore, since any two points in
 576 $B(p, R)$ can be connected by a unique minimal geodesic that remains entirely within $B(p, R)$, the
 577 geodesic ball $B(p, R)$ is geodesically convex in $\mathbb{S}^n$ for all radius $R < \pi/2\sqrt{K}$. This ensures that
 578 $\text{CAT}(K)$ condition preserves the strict convexity.

579 Given that $\text{diam}(\text{supp}(\nu)) < \pi/2\sqrt{K}$, for any geodesic $t \mapsto \gamma(t)$ connecting two distinct points
 580 $m_1, m_2 \in \mathcal{M}$, the Fréchet functional satisfies

$$F(\gamma(t)) < (1-t)F(m_1) + tF(m_2),$$

581 for all $t \in (0, 1)$, provided $m_1 \neq m_2$. Here, strict convexity of $F(x)$ ensures that any local minimum
 582 is a global minimum, and further, that such a minimum is unique within the convex neighborhood. $\square$

583 B.2 Proofs for Section 3.2

584 *Proof for Theorem 1.* Define the population Fréchet functional $F(z)$ and empirical Fréchet functional
585 $F_n(z)$ as follows.

$$F(z) := \mathbb{E}[d^2(Y, m)],$$

$$F_n(z) := \frac{1}{n} \sum_{i=1}^n d^2(Y_i, z).$$

586 By definition,

$$\mu = \arg \min_{z \in \mathcal{M}} F(z),$$

$$\hat{\mu}_n = \arg \min_{z \in \mathcal{M}} F_n(z).$$

587 Assume that μ is unique, which holds if $\text{diam}(\mathcal{M}) < \pi/2\sqrt{K}$ when $K > 0$ or automatically if
588 $K \leq 0$, from Lemmas 2, 3 and Propositions 1, 4.

589 A key geometric fact in $\text{CAT}(K)$ spaces is that the map

$$z \mapsto \mathbb{E}[d^2(Y, z)] = F(z)$$

590 is α -strongly geodesically convex around μ , provided $\text{diam}(\mathcal{M})$ is small enough. Concretely, there
591 exists a constant

$$\alpha = \alpha(K, D) > 0,$$

592 such that for every $z \in \mathcal{M}$,

$$F(z) - F(\mu) \geq \alpha d^2(z, \mu).$$

593 A fully explicit formula for $\alpha(K, D)$ can be extracted from standard $\text{CAT}(K)$ lemmas.

- 594 • If $K \leq 0$, one can take $\alpha(K, D) = \frac{1}{5}$. Indeed, $\text{CAT}(K)$ spaces are sometimes called
595 Hadamard spaces, for which $d^2(y, \cdot)$ is 1-convex along geodesics.
- 596 • If $K > 0$ but $\text{diam}(\mathcal{M}) = D < \pi/2\sqrt{K}$, one obtains an explicit lower bound

$$\alpha(K, D) \geq \frac{\sin(2\sqrt{K}R)}{2R},$$

597 where $R = D/2$. One often sees, for example,

$$\alpha(K, D) = \frac{2}{\pi} \sqrt{K} \sin\left(\frac{\pi}{2} - \sqrt{K}D\right).$$

598 Since $\hat{\mu}_n$ is the minimizer of F_n , one can obtain

$$F_n(\hat{\mu}_n) \leq F_n(\mu).$$

599 Here, rewriting $F_n = F_n - F + F$,

$$\begin{aligned} F_n(\hat{\mu}_n) - F_n(\mu) &= \{F_n(\hat{\mu}_n) - F(\hat{\mu}_n)\} - \{F_n(\mu) - F(\mu)\} + \{F(\mu_n) - F(\mu)\} \\ &\leq 0, \\ F(\hat{\mu}_n) - F(\mu) &\leq \{F_n(\mu) - F(\mu)\} - \{F_n(\hat{\mu}_n) - F(\hat{\mu}_n)\} \\ &\leq |F_n(\mu) - F(\mu)| + |F_n(\hat{\mu}_n) - F(\hat{\mu}_n)| \\ &\leq 2 \sup_{z \in \mathcal{M}} |F_n(z) - F(z)|. \end{aligned}$$

600 On the other hand, by the strong convexity of $F(z)$,

$$F(\hat{\mu}_n) - F(\mu) \geq \alpha(K, D) d^2(\hat{\mu}_n, \mu).$$

601 Therefore, by combining them, if $d(\hat{\mu}_n, \mu) \geq \epsilon$, then

$$\begin{aligned} \alpha(K, D)\epsilon^2 &\leq F(\hat{\mu}_n) - F(\mu) \\ &\leq 2 \sup_{z \in \mathcal{M}} |F_n(z) - F(z)|. \end{aligned}$$

602 Hence,

$$\{d(\hat{\mu}_n, \mu) \geq \epsilon\} \subseteq \left\{ \sup_{z \in \mathcal{M}} |F_n(z) - F(z)| \geq \frac{\alpha(K, D)}{2} \epsilon^2 \right\},$$

603 and

$$\mathbb{P}[d(\hat{\mu}_n, \mu) \geq \epsilon] \leq \mathbb{P}\left[\sup_{z \in \mathcal{M}} |F_n(z) - F(z)| \geq \frac{\alpha(K, D)}{2} \epsilon^2\right].$$

604 So, it suffices to control $\sup_{z \in \mathcal{M}} |F_n(z) - F(z)|$ by an exponential tail.

605 Recall that

$$F_n(z) - F(z) = \frac{1}{n} \sum_{i=1}^n \{d^2(Y_i, z) - \mathbb{E}[d^2(Y, z)]\}.$$

606 Define

$$X_i(z) = d^2(Y_i, z) - \mathbb{E}[d^2(Y, z)].$$

607 Then, $\mathbb{E}[X_i(z)] = 0$ and

$$F_n(z) - F(z) = \frac{1}{n} \sum_{i=1}^n X_i(z).$$

608 Because $\mathcal{M}$ has diameter $\text{diam}(\mathcal{M}) \leq D$, $d^2(\cdot, \cdot) \leq D^2$. Hence, for any z ,

$$X_i(z) \in [-D^2, D^2].$$

609 By Hoeffding's inequality, for a fixed z ,

$$\begin{aligned} \mathbb{P}[|F_n(z) - F(z)| \geq t] &= \mathbb{P}\left[\left|\sum_{i=1}^n X_i(z)\right| \geq nt\right] \\ &\leq 2 \exp\left(-\frac{nt^2}{2D^4}\right). \end{aligned}$$

610 Here, for every fixed z , one obtains a bound of the form

$$\mathbb{P}\left[\sup_{z \in \mathcal{M}} |F_n(z) - F(z)| \geq t\right] \leq c'_1 \exp(-c'_2 nt^2),$$

611 for constants $c'_1, c'_2 > 0$ depending on K, D and on the metric complexity of $\mathcal{M}$,

$$\begin{aligned} c'_1 &= 2 \left(\frac{\alpha(K, D)D}{\delta} \right)^m, \\ c'_2 &= \frac{\alpha(K, D)}{8D^2}, \end{aligned}$$

612 that are from standard references in manifold-valued statistics.

613 Putting it all together,

$$\begin{aligned} \mathbb{P}[d(\hat{\mu}_n, \mu) \geq \epsilon] &\leq \mathbb{P}\left[\sup_{z \in \mathcal{M}} |F_n(z) - F(z)| \geq \frac{\alpha(K, D)}{2} \epsilon^2\right] \\ &\leq c'_1 \exp\left\{-c'_2 n \left(\frac{\alpha(K, D)}{2} \epsilon^2\right)^2\right\}. \end{aligned}$$

614 This concludes the required proof. □

615 *Proof for Proposition 2.* By Theorem 1, there exist positive constants $c_1 = c_1(K, D)$ and $c_2 =$
 616 $c_2(K, D)$, such that for every $\epsilon > 0$,

$$\mathbb{P}[d(\hat{\mu}_n, \mu) > \epsilon] \leq c_1 \exp(-c_2 n \epsilon^2).$$

617 For any nonnegative random variable Z and any $p > 1$, one has the standard identity

$$\mathbb{E}[Z^p] = \int_0^\infty p \epsilon^{p-1} \mathbb{P}(Z > \epsilon) d\epsilon.$$

618 This follows from writing $\mathbb{E}[Z^p] = \int_0^\infty p \epsilon^{p-1} \mathbb{1}(Z > \epsilon) d\epsilon$ and exchanging expectation and integral.

619 Applying this to $Z = d(\hat{\mu}_n, \mu)$,

$$\mathbb{E}[d^p(\hat{\mu}_n, \mu)] = \int_0^\infty p \epsilon^{p-1} \mathbb{P}[d(\hat{\mu}_n, \mu) > \epsilon] d\epsilon.$$

620 Therefore,

$$\begin{aligned} \mathbb{E}[d^p(\hat{\mu}_n, \mu)] &\leq \int_0^\infty p \epsilon^{p-1} [c_1 \exp(-c_2 n \epsilon^2)] d\epsilon \\ &= c_1 \int_0^\infty p \epsilon^{p-1} \exp(-c_2 n \epsilon^2) d\epsilon. \end{aligned}$$

621 Let $u = \sqrt{n} \epsilon$. Then, $\epsilon = u/\sqrt{n}$ and $d\epsilon = \frac{1}{\sqrt{n}} du$. Also,

$$\begin{aligned} \epsilon^{p-1} &= \left(\frac{u}{\sqrt{n}}\right)^{p-1} = n^{-(p-1)/2} u^{p-1}, \\ \exp(-c_2 n \epsilon^2) &= \exp(-c_2 u^2). \end{aligned}$$

622 So,

$$\begin{aligned} \int_0^\infty \epsilon^{p-1} \exp(-c_2 n \epsilon^2) d\epsilon &= \int_0^\infty n^{-(p-1)/2} u^{p-1} \exp(-c_2 u^2) \frac{1}{\sqrt{n}} du \\ &= n^{-\frac{p-1}{2}} n^{-\frac{1}{2}} \int_0^\infty u^{p-1} \exp(-c_2 u^2) du \\ &= n^{-\frac{p}{2}} \int_0^\infty u^{p-1} \exp(-c_2 u^2) du. \end{aligned}$$

623 Now, evaluate $\int_0^\infty u^{p-1} \exp(-c_2 u^2) du$. This is a known integral that can be expressed via the
 624 Gamma function. Indeed,

$$\int_0^\infty u^{p-1} \exp(-c_2 u^2) du = \frac{1}{2} c_2^{-\frac{p}{2}} \Gamma\left(\frac{p}{2}\right),$$

625 and

$$\int_0^\infty \epsilon^{p-1} \exp(-c_2 n \epsilon^2) d\epsilon = n^{-\frac{p}{2}} \left[\frac{1}{2} c_2^{-\frac{p}{2}} \Gamma\left(\frac{p}{2}\right) \right].$$

626 Therefore,

$$\mathbb{E}[d^p(\hat{\mu}_n, \mu)] \leq c_1 p \left\{ n^{-\frac{p}{2}} \left[\frac{1}{2} c_2^{-\frac{p}{2}} \Gamma\left(\frac{p}{2}\right) \right] \right\}.$$

627 Collecting constants and it gives the proof. □

628 *Proof for Theorem 2.* Fix a point $x \in \mathbb{R}^d$. Define the weighted empirical measure of Y given x as

$$\nu_{n,x} := \sum_{i=1}^n w_{n,i}(x) \delta_{Y_i},$$

629 where δ_Y denotes the Dirac measure at Y . Because $\sum_{i=1}^n w_{n,i}(x) = 1$, this is indeed a probability
 630 measure on $\mathcal{M}$. Similarly, let ν_x be the true conditional distribution of Y given $X = x$ as

$$\nu_x := \mathbb{P}[Y \in A \mid X = x].$$

631 for Borel sets $A \subset \mathcal{M}$. Then, observe that the estimator $\hat{\mu}_n^*(x)$ can be written as

$$\begin{aligned}\hat{\mu}_n^*(x) &= \arg \min_{z \in \mathcal{M}} \sum_{i=1}^n w_{n,i}(x) d^2(Y_i, z) \\ &= \arg \min_{z \in \mathcal{M}} \int_{-\infty}^{+\infty} d^2(y, z) d\nu_{n,x}(y).\end{aligned}$$

632 That is, $\hat{\mu}_n^*(x)$ is precisely the Fréchet mean of the measure $\nu_{n,x}$. Meanwhile, $\mu^*(x)$ is the Fréchet
633 mean of ν_x :

$$\mu^*(x) = \arg \min_{z \in \mathcal{M}} \int_{-\infty}^{+\infty} d^2(y, z) d\nu_x(y).$$

634 Hence, the problem reduces to showing that as $n \rightarrow +\infty$, $\nu_{n,x}$ converges to ν_x in a sense strong
635 enough to force their Fréchet means to converge.

636 From Assumption 1, one can expect that for any bounded function $f: \mathcal{M} \rightarrow \mathbb{R}$,

$$\int f d\nu_{n,x} = \sum_{i=1}^n w_{n,i}(x) f(Y_i) \xrightarrow[n \rightarrow \infty]{a.s.} \mathbb{E}[f(Y) \mid X = x] = \int f d\nu_x.$$

637 Thus, $\nu_{n,x}$ converges to ν_x in the weak topology on probability measures.

638 For each measure ν , define its Fréchet functional $F_\nu: \mathcal{M} \rightarrow \mathbb{R}$ by

$$F_\nu(z) := \int d^2(y, z) d\nu(y).$$

639 Here,

$$\begin{aligned}\hat{\mu}_n^*(x) &= \arg \min_{z \in \mathcal{M}} F_{\nu_{n,x}}(z), \\ \mu^*(x) &= \arg \min_{z \in \mathcal{M}} F_{\nu_x}(z).\end{aligned}$$

640 One want $F_{\nu_{n,x}} \rightarrow F_{\nu_x}$ in a suitable sense that implies arg min convergence. In fact, for pointwise
641 consistency, it suffices to show that for each $z \in \mathcal{M}$,

$$F_{\nu_{n,x}}(z) = \sum_{i=1}^n w_{n,i}(x) d^2(Y_i, z) \xrightarrow[n \rightarrow \infty]{a.s.} \int d^2(y, z) d\nu_x(y) = F_{\nu_x}(z).$$

642 By Assumption 1, this convergence holds for each $z \in \mathcal{M}$.

643 To pass from pointwise convergence of $F_{\nu_{n,x}}$ to convergence of the minimizers $\hat{\mu}_n^*(x) \rightarrow \mu^*(x)$,
644 one can rely on the strict geodesic convexity of $d^2(\cdot, \cdot)$ in a $\text{CAT}(K)$ space with small diameter.
645 Concretely, from earlier arguments, there is a constant $\alpha(K, D)$ such that

$$F_{\nu_x}(z) - F_{\nu_x}(\mu^*(x)) \geq \alpha(K, D) d^2(z, \mu^*(x)),$$

646 for all $z \in \mathcal{M}$. This follows from the strong geodesic convexity of $z \mapsto \int d^2(y, z) d\nu_x(y)$. Equiv-
647 alently, if z is ϵ -far from $\mu^*(x)$, then $F_{\nu_x}(z)$ exceeds the global minimum $F_{\nu_x}(\mu^*(x))$ at least
648 $\alpha(K, D)\epsilon^2$.

649 Now, let $\epsilon > 0$. Suppose, contrary to what one want, that

$$d(\hat{\mu}_n^*(x), \mu^*(x)) \geq \epsilon.$$

650 By $\text{CAT}(K)$ -convexity,

$$F_{\nu_x}(\hat{\mu}_n^*(x)) - F_{\nu_x}(\mu^*(x)) \geq \alpha(K, D)\epsilon^2.$$

651 On the other hand,

$$F_{\nu_x}(\hat{\mu}_n^*(x)) - F_{\nu_x}(\mu^*(x)) = \{F_{\nu_{n,x}}(\hat{\mu}_n^*(x)) - F_{\nu_{n,x}}(\mu^*(x))\} + (F_{\nu_x} - F_{\nu_{n,x}})(\hat{\mu}_n^*(x)) - (F_{\nu_x} - F_{\nu_{n,x}})(\mu^*(x)).$$

652 Since $\hat{\mu}_n^*(x)$ minimizes $F_{\nu_{n,x}}$,

$$F_{\nu_{n,x}}(\hat{\mu}_n^*(x)) \leq F_{\nu_{n,x}}(\mu^*(x)).$$

653 Thus,

$$F_{\nu_{n,x}}(\hat{\mu}_n^*(x)) - F_{\nu_x}(\mu^*(x)) \leq (F_{\nu_x} - F_{\nu_{n,x}})(\hat{\mu}_n^*(x)) - (F_{\nu_x} - F_{\nu_{n,x}})(\mu^*(x)).$$

654 Hence,

$$\alpha(K, D)\epsilon^2 \leq |(F_{\nu_x} - F_{\nu_{n,x}})(\hat{\mu}_n^*(x))| + |(F_{\nu_x} - F_{\nu_{n,x}})(\mu^*(x))|.$$

655 But as $n \rightarrow +\infty$,

$$F_{\nu_{n,x}}(z) \rightarrow F_{\nu_x}(z),$$

656 pointwise for each z , so the difference $|F_{\nu_x}(z) - F_{\nu_{n,x}}(z)| \rightarrow 0$. By dominated convergence theorem,

$$\sup_{z \in \{\hat{\mu}_n^*(x), \mu^*(x)\}} |F_{\nu_{n,x}}(z) - F_{\nu_x}(z)| \xrightarrow[n \rightarrow 0]{a.s.} 0.$$

657 Hence, for large n , the right-hand side in the above inequality is smaller than $\frac{1}{2}\alpha(K, D)\epsilon^2$, which is
658 incompatible. Thus, for large n ,

$$d(\hat{\mu}_n^*(x), \mu^*(x)) < \epsilon,$$

659 and

$$\hat{\mu}_n^*(x) \xrightarrow{a.s.} \mu^*(x).$$

660 This completes the proof of pointwise consistency. □

661 *Proof for Theorem 3.* For each x , define the empirical weighted measure as follows.

$$\nu_{n,x} := \sum_{i=1}^n w_{n,i}(x) \delta_{Y_i},$$

662 where δ_y is the Dirac measure at y . Then,

$$\hat{\mu}_n^*(x) = \arg \min_{z \in \mathcal{M}} \int d^2(y, z) d\nu_{n,x}(y).$$

663 Simultaneously, define the local population measure near x :

$$\pi_{n,x} := \frac{\mathbb{E} \left[W \left(\frac{\|x - X\|}{h_n} \right) \mathbb{1}(Y \in \cdot) \right]}{\mathbb{E} \left[W \left(\frac{\|x - X\|}{h_n} \right) \right]},$$

664 which is the ideal measure that the kernel weighting is trying to approximate. Then define the local
665 population Fréchet mean as

$$\tilde{\mu}_n^*(x) = \arg \min_{z \in \mathcal{M}} \int d^2(y, z) d\pi_{n,x}(y).$$

666 Here, $\tilde{\mu}_n^*(x)$ is the minimizer of the population version of the local kernel functional, and $\hat{\mu}_n^*(x)$ is
667 the minimizer of the empirical version. Then one can write

$$d(\hat{\mu}_n^*(x), \mu^*(x)) \leq d(\hat{\mu}_n^*(x), \tilde{\mu}_n^*(x)) + d(\tilde{\mu}_n^*(x), \mu^*(x)).$$

668 Squaring and taking expectation, and applying $2ab < a^2 + b^2$, one can get a bias-variance decompo-
669 sition:

$$\mathbb{E}[d^2(\hat{\mu}_n^*(x), \mu^*(x))] \leq 2\mathbb{E}[d^2(\hat{\mu}_n^*(x), \tilde{\mu}_n^*(x))] + 2d^2(\tilde{\mu}_n^*(x), \mu^*(x)).$$

670 The first term in the right-hand side is the variance term, capturing how the empirical local measure
671 $\nu_{n,x}$ fluctuates around $\pi_{n,x}$. The second term in the right-hand side is the bias term, capturing how
672 the local population mean $\tilde{\mu}_n^*(x)$ differs from $\mu^*(x)$.

673 Recall that in a CAT(K) space, of diameter $\text{diam}(\mathcal{M}) \leq D$, there is a strong geodesic convexity
674 constant $\alpha(K, D)$ such that

$$\int d^2(y, z) d\nu(y) - \int d^2(y, z^*) d\nu(z^*) \geq \alpha(K, D)d^2(z, z^*),$$

675 for all probability measures ν on $\mathcal{M}$, provided the measure is fully supported in a ball of diameter
 676 $\text{diam}(\mathcal{M}) \leq D$. Hence, for the local measure $\pi_{n,x}$,

$$\int d^2(y, \hat{\mu}_n^*(x)) d\pi_{n,x} - \int d^2(y, \tilde{\mu}_n^*(x)) d\pi_{n,x}(y) \geq \alpha(K, D) d^2(\hat{\mu}_n^*(x), \tilde{\mu}_n^*(x)).$$

677 Because $\hat{\mu}_n^*(x)$ minimizes $\int d^2(y, z) d\nu_{n,x}(y)$,

$$\int d^2(y, \hat{\mu}_n^*(x)) d\nu_{n,x}(y) \leq \int d^2(y, \tilde{\mu}_n^*(x)) d\nu_{n,x}(y).$$

678 By subtracting the corresponding population measure integrals,

$$[\nu_{n,x} - \pi_{n,x}] d^2(\cdot, \hat{\mu}_n^*(x)) - [\nu_{n,x} - \pi_{n,x}] d^2(\cdot, \tilde{\mu}_n^*(x)) \leq \int d^2(y, \tilde{\mu}_n^*(x)) d\pi_{n,x}(y) - \int d^2(y, \hat{\mu}_n^*(x)) d\pi_{n,x}(y) \\ \int d^2(y, \hat{\mu}_n^*(x)) d\pi_{n,x}(y) - \int d^2(y, \tilde{\mu}_n^*(x)) d\pi_{n,x}(y) \leq \Delta_n(x),$$

679 where

$$\Delta_n(x) := |[\nu_{n,x} - \pi_{n,x}] d^2(\cdot, \hat{\mu}_n^*(x))| + |[\nu_{n,x} - \pi_{n,x}] d^2(\cdot, \tilde{\mu}_n^*(x))|.$$

680 Combining with the strong convexity inequality,

$$\alpha(K, D) d^2(\hat{\mu}_n^*(x), \tilde{\mu}_n^*(x)) \leq \Delta_n(x) \\ d^2(\hat{\mu}_n^*(x), \tilde{\mu}_n^*(x)) \leq \frac{\Delta_n(x)}{\alpha(K, D)}.$$

681 Taking expectation with respect to the sample $\{(X_i, Y_i)\}_{i=1}^n$,

$$\mathbb{E}[d^2(\hat{\mu}_n^*(x), \tilde{\mu}_n^*(x))] \leq \frac{\mathbb{E}[\Delta_n(x)]}{\alpha(K, D)}.$$

682 Recall that

$$\Delta_n(x) = |[\nu_{n,x} - \pi_{n,x}] d^2(\cdot, \hat{\mu}_n^*(x))| + |[\nu_{n,x} - \pi_{n,x}] d^2(\cdot, \tilde{\mu}_n^*(x))| \\ = \left| \sum_{i=1}^n w_{n,i}(x) \{d^2(Y_i, \hat{\mu}_n^*(x)) - \mathbb{E}[d^2(Y, \tilde{\mu}_n^*(x) | X \approx x)]\} \right| \\ + \left| \sum_{i=1}^n w_{n,i}(x) \{d^2(Y_i, \hat{\mu}_n^*(x)) - \mathbb{E}[d^2(Y, \tilde{\mu}_n^*(x) | X \approx x)]\} \right|.$$

683 Since $\hat{\mu}_n^*$ itself depends on the sample, a straightforward application of Hoeffding's inequality is
 684 tricky. However, one can use Efron–Stein or Bennett-type inequalities for U-statistics, or the bounded
 685 differences approach, carefully analyzing how a single Y_i affects $\hat{\mu}_n^*$. Such arguments appear in
 686 standard references on manifold-valued kernel regression. Thus, one can obtain

$$\mathbb{E}[\Delta_n(x)] = O\left((nh_n^d)^{-1/2}\right).$$

687 Hence,

$$\mathbb{E}[d^2(\hat{\mu}_n^*(x), \tilde{\mu}_n^*(x))] \leq \frac{C_{\text{var}}}{\alpha(K, D)} (nh_n^d)^{-1/2}.$$

688 where C_{var} is a constant depending on the kernel shape, the distribution of (X, Y) near x and the
 689 geometry constants (K, D) .

690 Next, recall that

$$\tilde{\mu}_n^*(x) = \arg \min_{z \in \mathcal{M}} \int d^2(y, z) d\pi_{n,x}(y), \\ \mu^*(x) = \arg \min_{z \in \mathcal{M}} \int d^2(y, z) d\nu_x(y),$$

691 where $\nu_x(\cdot) = \mathbb{P}[Y \in \cdot \mid X = x]$. As one move from $X = x$ to a local neighborhood $\{x' \mid$
692 $\|x - x'\| < O(h_n)\}$, it can be expected that $\mu_n^*(x)$ to approximate $\mu^*(x')$ for some $x' \approx x$. Then
693 $\mu^*(x')$ is close to $\mu^*(x)$ if μ^* is β -Hölder.

694 Because $\pi_{n,x}$ is essentially the distribution of $Y \mid X \in \{x' \mid \|x' - x\| < ch_n\}$, let x_n^* be some
695 effective point near x . Then by using smoothness or local Lipschitz condition on the conditional
696 distributions,

$$d(\tilde{\mu}_n^*(x), \mu^*(x')) \leq C_{\text{bias}}(h_n^\beta),$$

697 for some constant $C_{\text{bias}} > 0$. Then one adds

$$d(\mu^*(x'), \mu^*(x)) \leq L \cdot \|x' - x\| \approx Lh_n^\beta.$$

698 Hence,

$$d(\tilde{\mu}_n^*(x), \mu^*(x)) \leq d(\tilde{\mu}_n^*(x), \mu^*(x')) + d(\mu^*(x'), \mu^*(x)) = O(h_n^\beta),$$

699 and

$$d^2(\tilde{\mu}_n^*(x), \mu^*(x)) = O(h_n^{2\beta}).$$

700 Putting it all together in the bias–variance decomposition, it completes the required proof. $\square$

701 **B.3 Proofs for Section 3.3**

702 *Proof for Lemma 6* Let y' be a point on the geodesic segment $[xy]$ such that y' is very close to x .
 703 Similarly, pick z' on $[xz]$. So,

$$\begin{aligned} d(x, y') &= \delta, \\ d(x, z') &= \delta, \end{aligned}$$

704 for some $\delta > 0$. This triangle $\triangle xy'z'$ has perimeter $\leq d(x, y) + d(y, z) + d(z, x)$, which is assumed
 705 $\leq \pi/\sqrt{K}$ if $K > 0$. For δ small enough, the side lengths of $\triangle xy'z'$ are also $\leq \pi/\sqrt{K}$. By the
 706 CAT(K) definition,

$$d(y', z') \leq d_{\mathbb{M}_K}(\bar{y}', \bar{z}'),$$

707 and

$$\begin{aligned} d(x, y') &= d(\bar{x}, \bar{y}') = \delta, \\ d(x, z') &= d(\bar{x}, \bar{z}') = \delta. \end{aligned}$$

708 The triangle $\triangle \bar{x}\bar{y}'\bar{z}'$ is in the same model plane as $\triangle \bar{x}\bar{y}\bar{z}$, but its typically much smaller near $\bar{x}$.

709 By definition of the Alexandrov angle,

$$\angle_x(y, z) = \lim_{\delta \rightarrow 0} \angle_x^{(\text{sec})}(y', z'),$$

710 where $\angle_x^{(\text{sec})}(y', z')$ is the secular angle of $\triangle xy'z'$ at x . Equivalently, it is the Euclidean angle
 711 $\angle_{\bar{x}}(\bar{y}', \bar{z}')$ in the comparison triangle $\triangle \bar{x}\bar{y}'\bar{z}'$. Thus,

$$\angle_x(y, z) = \lim_{\delta \rightarrow 0} \angle_{\bar{x}}(\bar{y}', \bar{z}').$$

712 One also have the angle $\angle_{\bar{x}}(\bar{y}, \bar{z})$ in the large triangle $\triangle \bar{x}\bar{y}\bar{z}$, and want to show

$$\angle_{\bar{x}}(\bar{y}', \bar{z}') \leq \angle_{\bar{x}}(\bar{y}, \bar{z}).$$

713 for each small δ from which it will follow in the limit that $\angle_x(y, z) \leq \angle_{\bar{x}}(\bar{y}, \bar{z})$.

714 The CAT(K) condition states that $\triangle xy'z'$ is no thicker than the model $\triangle \bar{x}\bar{y}'\bar{z}'$. More precisely, if
 715 one place $\triangle xy'z'$ and $\triangle \bar{x}\bar{y}'\bar{z}'$ side by side so that $x \leftrightarrow \bar{x}$, $y' \leftrightarrow \bar{y}'$, $z' \leftrightarrow \bar{z}'$ correspond, one have

$$d(y', z') \leq d_{\mathbb{M}_K}(\bar{y}', \bar{z}').$$

716 Meanwhile, $\triangle \bar{x}\bar{y}'\bar{z}' \subset \triangle \bar{x}\bar{y}\bar{z}$ or can be inscribed in it, with the property that as $y' \rightarrow x$ and $z' \rightarrow x$,
 717 the points $\bar{y}' \rightarrow \bar{x}$ and $\bar{z}' \rightarrow \bar{x}$.

718 Geometrically, on the model side, it is known (from classical geometry in constant curvature) that

$$\angle_{\bar{x}}(\bar{y}', \bar{z}') \leq \angle_{\bar{x}}(\bar{y}, \bar{z}). \quad (15)$$

719 This is because in a convex geometry (like a sphere of radius $1/\sqrt{K}$ or a Euclidean plane if $K = 0$),
 720 drawing smaller radii $\bar{x}\bar{y}'$ and $\bar{x}\bar{z}'$ inside the bigger radii $\bar{x}\bar{y}$ and $\bar{x}\bar{z}$ yields smaller or equal angles
 721 from the center $\bar{x}$.

722 More precisely, if one revolve the segment $\bar{y}'\bar{z}'$ about $\bar{x}$ within the triangle $\triangle \bar{x}\bar{y}\bar{z}$, the angle $\angle_{\bar{x}}(\bar{y}', \bar{z}')$
 723 cannot exceed $\angle_{\bar{x}}(\bar{y}, \bar{z})$.

724 One thus have, for each small $\delta > 0$,

$$\angle_{\bar{x}}(\bar{y}', \bar{z}') \leq \angle_{\bar{x}}(\bar{y}, \bar{z}).$$

725 By the definition,

$$\angle_x(y, z) = \lim_{\delta \rightarrow 0} \angle_{\bar{x}}(\bar{y}', \bar{z}') \leq \angle_{\bar{x}}(\bar{y}, \bar{z}).$$

726 This completes the proof. Thus the angle at x in the real triangle $\triangle xyz$ is bounded above by the
 727 corresponding angle at $\bar{x}$ in the comparison triangle $\triangle \bar{x}\bar{y}\bar{z}$. $\square$

728 *Proof for Lemma 7.* Let $\triangle pqr \subset \mathcal{M}$ have side lengths

$$a = d(p, q), \quad b = d(q, r), \quad c = d(r, p),$$

729 and let $\angle_p(q, r)$ denote the Alexandrov angle at p . Similarly, let $\triangle p'q'r'$ have side lengths

$$a' = d(p', q'), \quad b' = d(q', r'), \quad c' = d(r', p'),$$

730 with angle $\angle_{p'}(q', r')$.

731 Assume that both triangles have perimeter $\leq \pi/\sqrt{K}$ if $K > 0$, ensuring they can be compared to

732 triangles in the simply connected model space of curvature K (sphere of radius $1/\sqrt{K}$ if $K > 0$,

733 Euclidean plane if $K = 0$, or hyperbolic plane if $K < 0$). Then, the goal is to show that

$$|\angle_p(q, r) - \angle_{p'}(q', r')| \leq C [d(p, p') + d(q, q') + d(r, r')],$$

734 for some constant C depending on $\alpha(K, D)$ or directly $\pi/\sqrt{K}$.

735 From the triangle inequality, one get for instance

$$\begin{aligned} |a - a'| &= |d(p, q) - d(p', q')| \\ &\leq d(p, p') + d(q, q'), \end{aligned}$$

736 and similarly,

$$\begin{aligned} |b - b'| &\leq d(q, q') + d(r, r'), \\ |c - c'| &\leq d(r, r') + d(p, p'). \end{aligned}$$

737 Hence, each difference in corresponding side lengths is at most

$$\max\{|a - a'|, |b - b'|, |c - c'|\} \leq d(p, p') + d(q, q') + d(r, r') =: \delta_{pp'qq'rr'}.$$

738 Then,

$$|a - a'| \leq \delta_{pp'qq'rr'}, \quad |b - b'| \leq \delta_{pp'qq'rr'}, \quad |c - c'| \leq \delta_{pp'qq'rr'}.$$

739 In classical geometry of constant curvature K (sphere, Euclidean plane, and hyperbolic plane),

740 the side lengths (a, b, c) uniquely determine the shape of a triangle (up to rigid motion) provided

741 a, b, c satisfy the triangle inequality. The angle $\eta := \angle_p(q, r)$ (or its model-space counterpart $\tilde{\eta}$) is a

742 continuous function of (a, b, c) .

743 • If $K = 0$ (Euclidean), one have the law of cosines

$$c^2 = a^2 + b^2 - 2ab \cos(\eta),$$

744 so

$$\cos(\eta) = \frac{a^2 + b^2 + c^2}{2ab}.$$

745 This is a rational, continuous function of (a, b, c) .

746 • If $K > 0$ (spherical), the spherical law of cosines yield

$$\cos(\sqrt{K}c) = \cos(\sqrt{K}a) \cos(\sqrt{K}b) + \sin(\sqrt{K}a) \sin(\sqrt{K}a) \sin(\sqrt{K}b) \cos(\eta).$$

747 • If $K < 0$ (hyperbolic), one have similar hyperbolic law of cosines with cosh and sinh.

$$\cosh(c/K) = \cosh(a/K) \cosh(b/K) - \sinh(a/K) \sinh(b/K) \cos(\eta).$$

748 In each case, as long as $a, b, c \leq \pi/\sqrt{|K|}$, one remain in a region where the side-length-angle

749 relation is well-defined and continuously differentiable. Then, there exists a function

$$F: \{(a, b, c)\} \subset \mathbb{R}_{>0}^3 \rightarrow [0, \pi],$$

750 so that if $\triangle xyz$ in the model space has sides (a, b, c) , then the angle at x is $F(a, b, c)$. Moreover,
 751 F is Lipschitz continuous on the domain $\{(a, b, c) \mid a + b + c \leq \pi/\sqrt{K}\}$. Hence, if (a, b, c) and
 752 (a', b', c') are close in $\mathbb{R}^3$, then

$$|F(a, b, c) - F(a', b', c')| \leq K_0 (|a - a'| + |b - b'| + |c - c'|),$$

753 for some constant K_0 depending only on $\max(a, b, c) \leq \pi/\sqrt{K}$.

754 Now connect the actual angles $\angle_p(q, r)$, $\angle_{p'}(q', r')$ in $\text{CAT}(K)$ to their comparison angles $\bar{\alpha}$, $\bar{\alpha}'$ in
 755 the model space. For $\triangle pqr \subset M$, choose the comparison triangle $\triangle \bar{p}\bar{q}\bar{r} \subset \bar{M}$ in the model space of
 756 curvature K , with side lengths $\bar{p}\bar{q} = a$, $\bar{q}\bar{r} = b$, $\bar{r}\bar{p} = c$. Let $\bar{\eta} = \angle_{\bar{p}}(\bar{q}, \bar{r})$. For $\triangle p'q'r' \subset M$, choose
 757 $\triangle \bar{p}'\bar{q}'\bar{r}' \subset \bar{M}$ similarly with side lengths a' , b' , c' . Let $\bar{\eta}' = \angle_{\bar{p}'}(\bar{q}', \bar{r}')$.

758 By Lemma 6 in $\text{CAT}(K)$:

$$\begin{aligned} \angle_p(q, r) &\leq \bar{\eta}, \\ \angle_{p'}(q', r') &\leq \bar{\eta}'. \end{aligned}$$

759 Symmetrically reversing the roles, one also get

$$\bar{\eta} \leq \angle_p(q, r).$$

760 Here, $\angle_p(q, r) \approx \bar{\eta}$ and $\angle_{p'}(q', r') \approx \bar{\eta}'$. Hence

$$|\angle_p(q, r) - \angle_{p'}(q', r')| \leq |\bar{\alpha} - \bar{\eta}'| + |\angle_p(q, r) - \bar{\eta}| + |\angle_{p'}(q', r') - \bar{\eta}'|.$$

761 But each difference $|\angle_p(q, r) - \bar{\eta}|$ is known to be small by the usual $\text{CAT}(K)$ thin triangle property.
 762 Specifically, if the perimeter is $\leq \pi/\sqrt{K}$, the difference $\angle_p(q, r) - \bar{\eta}$ can be bounded by a constant
 763 times the diameter of $\triangle pqr$; but that diameter is $\leq \max(a, b, c)$, already controlled.

764 In fact, in standard statements, one typically get an inequality of the form

$$|\angle_p(q, r) - \bar{\eta}| \leq \varepsilon_1(a, b, c) \quad \text{with } \varepsilon_1 \rightarrow 0 \text{ as } a, b, c \rightarrow 0,$$

765 and similarly for $\angle_{p'}(q', r')$. Since one are only after a linear bound in the final statement, it suffices
 766 that each difference is bounded by a universal constant (depending on $\pi/\sqrt{K}$). Thus, effectively

$$|\angle_p(q, r) - \angle_{p'}(q', r')| \leq 2(\text{const}) + |\bar{\eta} - \bar{\eta}'|.$$

767 Hence collecting all,

$$|\angle_p(q, r) - \angle_{p'}(q', r')| \leq C_1 + C_2 \Delta$$

768 for constants C_1 and C_2 . In typical statements of the lemma, one either arranges that Δ is small so
 769 that the additive constant C_1 is overshadowed, or uses a slightly refined thinness difference argument
 770 to show $\angle_p(q, r)$ and $\bar{\eta}$ differ by $\leq \tilde{C} \cdot \Delta$. In either case, one get a final bound of the form

$$|\angle_p(q, r) - \angle_{p'}(q', r')| \leq C \Delta = C(d(p, p') + d(q, q') + d(r, r')).$$

771 This completes the proof. □

772 *Proof for Proposition 3.* First, from the geodesic convexity, if ν_x and $\nu_{x'}$ are close in distribution,
 773 then

$$d(\mu^*(x), \mu^*(x')) = C''\epsilon,$$

774 for some constant C'' depending on $\alpha(K, D)$ and distributional assumptions (e.g. sub-Gaussianity or
 775 bounded diameter ensuring all integrals are finite).

776 Compare angles $\angle_{\mu^*(x)}(u, v)$ and $\angle_{\mu^*(x')}(u, v)$. Let $[\mu^*(x), u]$ be the geodesic from $\mu^*(x)$ to u ,
 777 $[\mu^*(x'), u]$ be the geodesic from $\mu^*(x')$ to u , and similarly for $[\mu^*(x), v]$ and $[\mu^*(x'), v]$. Consider
 778 two triangles $\triangle(\mu^*(x), u, \mu^*(x'))$ and $\triangle(\mu^*(x), v, \mu^*(x'))$. Observe that $\text{diam}(\mathcal{M}) \leq D$, so if
 779 $\mu^*(x)$ and $\mu^*(x')$ are also $\leq O(\epsilon)$ apart, then each of these triangles has perimeter $2D + O(\epsilon)$. If
 780 $K > 0$, $2D + O(\epsilon) < \pi/(\sqrt{K})$ by the initial assumption $D < \frac{\pi}{2\sqrt{K}}$ and ϵ small enough. Hence,
 781 each triangle is validly contained in a region where one can apply $\text{CAT}(K)$ angle comparisons (and
 782 the model-space comparison).

783 Let

$$p = \mu^*(x), \quad q = u, \quad r = \mu^*(x').$$

784 and

$$p' = \mu^*(x'), \quad q' = u, \quad r' = \mu^*(x).$$

785 Then the pair $\triangle pqr$ and $\triangle p'q'r'$ have corresponding points:

$$p \leftrightarrow p', \quad q \leftrightarrow q', \quad r \leftrightarrow r'.$$

786 Notice that $q = q'$ is actually the same point u . The sum of vertex perturbations is

$$\begin{aligned} d(p, p') + d(q, q') + d(r, r') &= d(\mu^*(x), \mu^*(x')) + 0 + d(\mu^*(x'), \mu^*(x)) \\ &= 2d(\mu^*(x), \mu^*(x')), \end{aligned}$$

787 and $d(\mu^*(x), \mu^*(x')) \leq C'' \epsilon$. By Lemma 7,

$$|\angle_p(q, r) - \angle_{p'}(q', r')| \leq C_1 [d(p, p') + d(q, q') + d(r, r')]$$

788 Hence

$$\begin{aligned} |\angle_{\mu^*(x)}(u, \mu^*(x')) - \angle_{\mu^*(x')}(u, \mu^*(x))| &\leq C_1 (2d(\mu^*(x), \mu^*(x'))) \\ &\leq 2C_1 C'' \epsilon. \end{aligned}$$

789 Similarly, for $\triangle \mu^*(x) v \mu^*(x')$, one get the same type of bound in terms of ϵ .

790 Recall that $\angle_{\mu^*(x)}(u, v)$ is the Alexandrov angle between geodesics $[\mu^*(x)u]$ and $[\mu^*(x)v]$. In a
791 CAT(K) space, the angle $\angle_{\mu^*(x)}(u, v)$ can be added or compared if we know angles involving a
792 third point $\mu^*(x')$. Thus,

$$|\angle_{\mu^*(x)}(u, v) - (\angle_{\mu^*(x)}(u, \mu^*(x')) + \angle_{\mu^*(x')}(u, v) - \pi)| \leq C_2 \cdot d(\mu^*(x), \mu^*(x')),$$

793 for some constant C_2 .

794 Putting all these small angle increments together, conclude that

$$|\angle_{\mu^*(x)}(u, v) - \angle_{\mu^*(x')}(u, v)| \leq C d(\mu^*(x), \mu^*(x')) = O(\epsilon).$$

795 Hence the angles at $\mu^*(x)$ versus $\mu^*(x')$ differ by a linear factor in ϵ . □

796 *Proof for Theorem 4.* From Proposition 3, if $\nu_x \approx \nu_{x'}$ (i.e. their distance is $\leq \delta$), then for any pair
797 (u, v) ,

$$|\angle_{\mu^*(x)}(u, v) - \angle_{\mu^*(x')}(u, v)| \leq C_1 \epsilon,$$

798 for some constant $C_1 > 0$. Hence for one pair of directions (u, v) , one get a linear-in- ϵ bound on
799 how much the angle can change.

800 Now consider not just one pair, but all pairs (u_i, u_j) with $1 \leq i < j \leq m$. But since each
801 $\angle_{\mu^*(x)}(u_i, u_j)$ is covered by the same result,

$$|\angle_{\mu^*(x)}(u_i, u_j) - \angle_{\mu^*(x')}(u_i, u_j)| \leq C_1 \epsilon,$$

802 for each pair (u_i, u_j) . Then the supremum over $i < j$ is also $\leq C_1 \epsilon$. In fact, it is not even needed a
803 union bound in probability sense, and each pair is bounded by the same linear factor $C_1 \epsilon$. Hence

$$\sup_{1 \leq i < j \leq m} |\angle_{\mu^*(x)}(u_i, u_j) - \angle_{\mu^*(x')}(u_i, u_j)| \leq C_1 \epsilon.$$

804 Thus one immediately extend from one pair to all $\binom{m}{2}$ pairs (u_i, u_j) .

805 In the hypothesis, it is typically stated that whenever $\|x - x'\|$ is small, then ν_x and $\nu_{x'}$ differ by
806 $\epsilon(\|x - x'\|)$. For instance, in a classical kernel or smoothing scenario, if $\|x - x'\| \leq \delta$, then

$$d_W(\nu_x, \nu_{x'}) \leq \epsilon(\delta)$$

807 Hence setting $\epsilon = \epsilon(\delta)$, for $\|x - x'\| \leq \delta$,

$$\sup_{1 \leq i < j \leq m} |\angle_{\mu^*(x)}(u_i, u_j) - \angle_{\mu^*(x')}(u_i, u_j)| \leq C_1 \epsilon(\delta).$$

808 Thus the angle difference is a function of δ . Hence define $C := C_1$ (it might also absorb small
809 distributional constants if needed), and putting it all together yields the proof. □

810 **B.4 Proofs for Section 3.4**

811 *Proof for Lemma 8.* In a smooth Riemannian manifold, for sufficiently close u and v , the unique
 812 geodesics $\gamma_u : [0, \|U\|] \rightarrow \mathcal{M}$ and $\gamma_v : [0, \|V\|] \rightarrow \mathcal{M}$ from z to u , respectively from z to v
 813 have well-defined initial velocity vectors at z . Let $\dot{\gamma}_u(0) \in T_z \mathcal{M}$ be the tangent vector to γ_u at z .
 814 By construction, this is precisely U if we identify $U \in T_z \mathcal{M}$ with the velocity vector in normal
 815 coordinates. Similarly, $\dot{\gamma}_v(0) = V \in T_z \mathcal{M}$.

816 In Riemannian geometry (without singularities around z), one then have:

$$\angle_z(u, v) = \angle(\dot{\gamma}_u(0), \dot{\gamma}_v(0)) = \cos^{-1} \left(\frac{g_z(\dot{\gamma}_u(0), \dot{\gamma}_v(0))}{\|\dot{\gamma}_u(0)\| \|\dot{\gamma}_v(0)\|} \right).$$

817 Here $g_z(\cdot, \cdot)$ is the Riemannian metric at z . In simpler notation, if one identify $\dot{\gamma}_u(0) = U$ and
 818 $\dot{\gamma}_v(0) = V$, then

$$\angle_z(u, v) = \cos^{-1} \left(\frac{g_z(U, V)}{\sqrt{g_z(U, U) g_z(V, V)}} \right).$$

819 Use a geodesic coordinate system $\Phi : T_z \mathcal{M} \supset B_\delta(0) \rightarrow \mathcal{M}$ around z with $\Phi(0) = z$ and $d\Phi|_0 = \text{Id}$.
 820 Concretely, $\Phi(U) = \exp_z(U)$. In these coordinates, the metric $g_{ij}(X)$ at a point X in a small ball
 821 around $0 \in T_z \mathcal{M}$ has the well-known expansions:

$$g_{ij}(X) = \delta_{ij} - \frac{1}{3} R_{ikj\ell}(0) X^k X^\ell + O(\|X\|^3),$$

822 where $R_{ikj\ell}$ is the Riemann curvature tensor at z . The $-\frac{1}{3}$ factor is a standard convention from
 823 normal coordinate expansions; the main point is that the first non-trivial corrections appear at second
 824 order in $\|X\|$.

825 Hence, for vectors $U, V \in T_z \mathcal{M}$ with small norms, the inner product in the manifold at z is

$$g_z(U, V) = \delta_{ij} U^i V^j - \frac{1}{3} \sum_{k, \ell} \left(\frac{1}{2} R_{ikj\ell}(0) \right) \dots + O(\|U\| \|V\| \max(\|U\|, \|V\|)).$$

826 In simpler notation:

$$g_z(U, V) = \langle U, V \rangle_{\text{Eucl}} + O(\|U\| \|V\| \max(\|U\|, \|V\|)).$$

827 From the above expansions,

$$\sqrt{g_z(U, U)} = \|U\|_{\text{Eucl}} [1 + O(\|U\|^2)]^{1/2} = \|U\| + O(\|U\|^3).$$

828 Similarly for $\|V\|$. In addition,

$$g_z(U, V) = \langle U, V \rangle_{\text{Eucl}} + O(\|U\| \|V\| \max(\|U\|, \|V\|)).$$

829 Thus

$$\frac{g_z(U, V)}{\sqrt{g_z(U, U) g_z(V, V)}} = \frac{\langle U, V \rangle}{\|U\| \|V\|} + O(\|U\|^2 + \|V\|^2),$$

830 since each correction is second-order in $\|U\|$ or $\|V\|$. Moreover,

$$\angle_z(u, v) = \cos^{-1} \left(\frac{g_z(U, V)}{\sqrt{g_z(U, U) g_z(V, V)}} \right) = \cos^{-1} \left(\frac{\langle U, V \rangle}{\|U\| \|V\|} + O(\|U\|^2 + \|V\|^2) \right).$$

831 When $\theta_0 = \angle_0(U, V)$ denotes the Euclidean angle in the tangent space,

$$\cos(\theta_0) = \frac{\langle U, V \rangle}{\|U\| \|V\|}.$$

832 Then

$$\cos(\angle_z(u, v)) = \cos(\theta_0) + O(\|U\|^2 + \|V\|^2).$$

833 Since $\cos$ is locally invertible around angles not equal to $0, \pi$ (and we assume θ_0 is not degenerate or
834 extremely close to π for typical use), a standard expansion yields:

$$\angle_z(u, v) = \theta_0 + O(\|U\|^2 + \|V\|^2).$$

835 Concretely, if $\theta_1 = \theta_0 + \delta$ satisfies $\cos(\theta_1) = \cos(\theta_0) + \eta$, then $\delta = O(\eta)$ for small η . Here,
836 $\eta = O(\|U\|^2 + \|V\|^2)$.

837 Hence,

$$\angle_z(u, v) = \theta_0 + O(\|U\|^2 + \|V\|^2),$$

838 where $\theta_0 = \angle_0(U, V)$ is the Euclidean angle of U and V in $T_z M$. This completes the proof. $\square$

839 *Proof for Proposition 4.* Let $\gamma(t)$ be a geodesic in (M, g) with $\gamma(0) = \mu^*$ and $\dot{\gamma}(0) = v$. Consider
840 $F(\gamma(t))$. Then

$$\begin{aligned} \frac{d}{dt} F(\gamma(t)) \Big|_{t=0} &= \frac{d}{dt} \int d^2(y, \gamma(t)) d\nu(y) \Big|_{t=0} \\ &= \int \frac{d}{dt} d^2(y, \gamma(t)) \Big|_{t=0} d\mu(y). \end{aligned}$$

841 By standard Riemannian geometry formulas, if $\sigma(s)$ is the geodesic $[y, \gamma(t)]$, then

$$\frac{d}{dt} d^2(y, \gamma(t)) = 2 d(y, \gamma(t)) \left\langle \dot{\gamma}(t), \dot{\sigma}(0) \right\rangle_{g_{\gamma(t)}}.$$

842 At $t = 0$, since $\gamma(0) = \mu^*$, one interpret $\dot{\sigma}(0)$ as the initial velocity from μ^* toward y . If μ^* is a
843 minimizer, the directional derivative must vanish for all directions y . Formally, this implies

$$\nabla F(\gamma^*) = 0.$$

844 Hence the first-order term in the expansion of $F(z)$ around $z = \mu^*$ vanishes.

845 Next, examine the second derivative (or Hessian) of F at γ^* .

$$\text{Hess}_z(F)(v, v) = \frac{d^2}{dt^2} F(\exp_z(t v)) \Big|_{t=0}.$$

846 When $z = \mu^*$, and μ^* is the unique minimizer, these second derivatives measure how strongly F
847 curves upward around μ^* .

848 In fact, the Gauss–Manasse–Busemann formula for second variation of distance shows that

$$H_{\mu^*}(F)(v, v) = \int H_{\mu^*}[d^2(y, \cdot)](v, v) d\mu(y).$$

849 Each term $H_{\mu^*}[d^2(y, \cdot)](v, v)$ can be computed from the second variation of $\rho(\mu^*, y) = d(\mu^*, y)$.
850 In standard curvature conditions (especially nonpositive curvature or small diameter in positive
851 curvature), this Hessian is positive semidefinite, ensuring local convexity around μ^* . If $\text{CAT}(0)$
852 or if $\text{diam} < \pi/(2\sqrt{K})$ in $\text{CAT}(K)$, then $d^2(y, \cdot)$ is geodesically convex with a definite strong
853 convexity modulus $\alpha > 0$. Integrating preserves that positivity, giving $H_{\mu^*}(F) \succeq 0$. Hence there is a
854 well-defined linear operator H_{μ^*} on $T_{\mu^*} M$ representing $H_{\mu^*}(F)$.

855 Because F is at least C^2 , one can write the remainder $R(v)$ in a standard Taylor expansion form:

$$R(v) = O(\|v\|^3) \quad \text{as } v \rightarrow 0.$$

856 Concretely, one can show this by analyzing the third derivative of F in normal coordinates:

$$\frac{d^3}{dt^3} F(\exp_{\mu^*}(t v))$$

857 remains bounded as $t \rightarrow 0$, so the third-order term is well-defined.

858 Hence the local expansion is

$$F(\exp_{\mu^*}(v)) = F(\mu^*) + \underbrace{\langle \nabla F(\mu^*), v \rangle}_{=0} + \frac{1}{2} \langle H_{\mu^*} v, v \rangle + R(v), \quad R(v) = O(\|v\|^3).$$

859 That is precisely the jet expansion for the Fréchet functional around μ^* . $\square$

860 B.5 Proofs for Section 3.5

861 *Proof for Proposition 5.* From the local Riemannian (or CAT(K)) law of cosines in $\Delta_{\mu^*} y z$:

$$d^2(y, z) = d^2(y, \mu^*) + d^2(z, \mu^*) - 2 d(y, \mu^*) d(\mu^*, z) \cos(\angle_{\mu^*}(y, z)).$$

862 Rewriting as

$$d^2(y, z) - d^2(y, \mu^*) = d^2(z, \mu^*) - 2 d(y, \mu^*) d(\mu^*, z) \cos(\angle_{\mu^*}(y, z)).$$

863 Here, let

$$\Delta_{\text{dist}}(y, z, \mu^*) := d^2(\mu^*, z) - 2 d(y, \mu^*) d(\mu^*, z) \cos(\angle_{\mu^*}(y, z)).$$

864

$$\Delta_{\text{angle}}(y, z, \mu^*) := 2 d(y, \mu^*) d(z, \mu^*) [1 - \cos(\angle_{\mu^*}(y, z))].$$

865 Observe that

$$-2 d(y, \mu^*) d(\mu^*, z) \cos(\angle_{\mu^*}(y, z)) = [\Delta_{\text{dist}} - d^2(\mu^*, z)] - \Delta_{\text{angle}}.$$

866 and

$$d^2(y, z) = d^2(y, \mu^*) + \Delta_{\text{dist}}(y, z, \mu^*) + \Delta_{\text{angle}}(y, z, \mu^*).$$

867 So the desired identity is obtained. □

868 *Proof for Proposition 6.* Let

- 869 • $r_0 = d(\mu^*(x), u_0)$. (A constant for each x if u_0 is fixed.)
- 870 • $r(y) = d(\mu^*(x), y) = R_x(y)$. (A variable depending on y .)
- 871 • $\alpha(y) = d(u_0, y)$. Another side of the triangle.

872 Then from the local law of cosines,

$$r(y)^2 = r_0^2 + \alpha(y)^2 - 2 r_0 \alpha(y) \cos(\angle_{\mu^*(x)}(u_0, y)).$$

873 But $\angle_{\mu^*(x)}(u_0, y) = \phi_x(y)$. So

$$r(y)^2 = r_0^2 + \alpha(y)^2 - 2 r_0 \alpha(y) \cos(\phi_x(y)).$$

874 We write it as

$$\Psi_x(y) = r(y)^2 = r_0^2 + \alpha(y)^2 - 2 r_0 \alpha(y) \cos(\phi_x(y)).$$

875 Now, to link $\alpha(y) = d(u_0, y)$ with $r(y)$ and $\phi_x(y)$, we may do yet another small expansion or an
 876 additional law-of-cosines approach. If the manifold is small enough in diameter, we can treat $\alpha(y)$
 877 also as a function of $(r(y), \phi_x(y))$.

878 Also, let

$$\alpha(y)^2 = r_0^2 + r(y)^2 - 2 r_0 r(y) \cos(\angle_{u_0}(\mu^*(x), y)).$$

879 But $\angle_{u_0}(\mu^*(x), y)$ is not necessarily the same as $\phi_x(y)$. Then,

$$\alpha(y) = \alpha(r(y), \phi_x(y)) = r_0 + O(r(y))$$

880 plus terms involving $\phi_x(y)$. In a small neighborhood, these expansions typically become second-order
 881 in $\phi_x(y)$. Hence, $\alpha(y)$ is not an independent variable; it's determined once $\phi_x(y)$ and $r(y) = R_x(y)$
 882 are known.

883 In addition,

$$r(y)^2 = r_0^2 + \alpha(y)^2 - 2 r_0 \alpha(y) \cos(\phi_x(y)).$$

884 This yields a final expression of form

$$r(y)^2 = r_0^2 + \left(\text{some linear or quadratic function in } r(y) \right) + \left(\text{terms in } \phi_x(y) \right).$$

885 In short, the function $\Psi_x(y) = r(y)^2$ can be viewed as

$$\Psi_x(y) = \underbrace{f_{\text{radial}}(r(y))}_{\text{part ignoring angles}} + \underbrace{f_{\text{angle}}(r(y), \phi_x(y))}_{\text{angle corrections}},$$

886 where f_{angle} is typically second-order or cross-term in $\phi_x(y)$.

887 Consider

$$\mathbb{E}_{\nu_x}[\Psi_x(Y)] = \int r(y)^2 d\nu_x(y).$$

888 Let

889 • $\mathbb{E}_{\nu_x}[r(Y)]$ as some average radius.

890 • $\mathbb{E}_{\nu_x}[\phi_x(Y)]$ as average angle.

891 One obtains expansions, where

$$\Psi_x(Y) - r(y)^2 \Big|_{\phi_x(Y)=0}$$

892 is some cross or higher-order term in $\phi_x(Y)$.

893 Then,

$$\mathbb{E}[\Psi_x(Y)^2] = \int [r(y)^2]^2 d\nu_x(y).$$

894 Expanding $[r(y)^2]^2$ yields

$$[r(y)^2]^2 = r(y)^4 = \left(f_{\text{radial}}(r(y)) + f_{\text{angle}}(r(y), \phi_x(y)) \right)^2.$$

895 One obtains terms:

- 896 • $[f_{\text{radial}}(r)]^2$,
- 897 • cross terms $2 f_{\text{radial}}(r) f_{\text{angle}}(r, \phi)$,
- 898 • $[f_{\text{angle}}(r, \phi)]^2$.

899 By taking expectation,

$$\mathbb{E}[r(y)^4] = \mathbb{E}\left([f_{\text{radial}}(r)]^2\right) + 2\mathbb{E}\left(f_{\text{radial}}(r) f_{\text{angle}}(r, \phi)\right) + \mathbb{E}\left([f_{\text{angle}}(r, \phi)]^2\right).$$

900 Then, $\text{Var}[\Psi_x(Y)] = \mathbb{E}[\Psi_x(Y)^2] - (\mathbb{E}[\Psi_x(Y)])^2$ can be rearranged, grouping the radial part of the
901 variance from the angle cross terms:

$$\text{Var}[\Psi_x(Y)] = \text{Var}\left(\underbrace{f_{\text{radial}}(r(Y))}_{\text{like } r(Y)^2 \text{ ignoring angles}}\right) + \text{Cov}[\phi_x(Y), r(Y)^2] + (\text{smaller or higher-order expansions in } \phi_x(Y)).$$

902 Explicitly, let

$$A_x(Y) = f_{\text{radial}}(r(Y)) \quad (\text{often} = r(Y)^2)$$

903 ignoring angular corrections, and

$$B_x(Y) = f_{\text{angle}}(r(Y), \phi_x(Y)) \quad (\text{some function capturing dependence on angle } \phi_x(Y)).$$

904 Then

$$\Psi_x(Y) = A_x(Y) + B_x(Y)$$

905 Using

$$\text{Var}[A + B] = \text{Var}[A] + \text{Var}[B] + 2 \text{Cov}(A, B),$$

906 one have

$$\text{Var}[\Psi_x(Y)] = \text{Var}[A_x(Y)] + \text{Var}[B_x(Y)] + 2 \text{Cov}(A_x(Y), B_x(Y)).$$

907 If $B_x(Y)$ is small or mostly depends on $\phi_x(Y)$ with some bounding condition, one can inter-
908 pret $\text{Var}[B_x(Y)]$ and $\text{Cov}(A_x(Y), B_x(Y))$ as cross/higher-order expansions. Here, $\text{Var}[A_x(Y)]$
909 is the purely radial piece $\text{Var}[R_x(Y)^2]$. The cross terms or expansions in $\phi_x(Y)$ become
910 $\text{Cov}(\phi_x(Y), R_x(Y)^2)$. Hence we get the claimed partial decomposition. $\square$

911 C Additional Analysis on ϵ -Approximate CAT(K) Space

912 In comparison geometry framework, the theoretical statements are provided on the model space with
 913 constant curvature. In practice, however, real-world datasets may lie in spaces that only approximately
 914 satisfy the curvature conditions. Below we introduce an ϵ -approximate version of CAT(K) space,
 915 and derive perturbed versions of existence, uniqueness, and convexity-type results.

916 **Definition 9** (ϵ -Approximate CAT(K) Space). Let $\epsilon > 0$. A geodesic metric space $(\mathcal{M}, d)$ is said to
 917 be ϵ -approximate CAT(K) space if for every geodesic triangle $\triangle pqr$ of perimeter less than $2D_K$
 918 (where $D_K = \pi/\sqrt{K}$ if $K > 0$, otherwise $D_K = \infty$), and for any points $\bar{x}$ and $\bar{y}$ on the edges $[pq]$
 919 and $[qr]$, respectively, one has

$$d(x, y) \leq d_{\mathbb{M}_K^2}(\bar{x}, \bar{y}) + \epsilon, \quad (16)$$

920 where $\triangle \bar{p}\bar{q}\bar{r} \subset \mathbb{M}_K^2$ is the usual comparison triangle in the simply connected model space of constant
 921 curvature K .

922 This definition allows a small additive slack ϵ in the usual comparison inequality. When $\epsilon = 0$, we
 923 recover the standard definition of CAT(K).

924 **Theorem 5** (Approximate Geodesic Convexity of Squared Distance). Let $(\mathcal{M}, d)$ be an ϵ -approximate
 925 CAT(K) space with $K < 0$. Fix any $p \in \mathcal{M}$, and define $f(x) = d^2(p, x)$. Then, for any geodesic
 926 $\gamma: [0, 1] \rightarrow \mathcal{M}$,

$$f(\gamma(t)) \leq (1-t)f(\gamma(0)) + tf(\gamma(1)) + O(\epsilon D), \quad (17)$$

927 where D is the diameter of the relevant geodesic segment under consideration, or the whole space if
 928 bounded.

929 *Proof.* Let $\gamma: [0, 1] \rightarrow \mathcal{M}$ be a geodesic from $\gamma(0) = x$ to $\gamma(1) = y$. Define $\gamma(t)$ as the point at
 930 parameter t . We form a (possibly degenerate) triangle $\triangle pxy$ in $\mathcal{M}$. Then, $\triangle \bar{p}\bar{x}\bar{y}$ is the comparison
 931 triangle in the model space $\mathbb{M}_K^2$ that has side lengths

$$d_{\mathbb{M}_K^2}(\bar{p}, \bar{x}) = d(p, x), \quad d_{\mathbb{M}_K^2}(\bar{x}, \bar{y}) = d(x, y), \quad d_{\mathbb{M}_K^2}(\bar{y}, \bar{p}) = d(y, p).$$

932 Let $\bar{\gamma}(t)$ be the point on $[\bar{x}, \bar{y}] \subset \triangle \bar{p}\bar{x}\bar{y}$ at fraction t . Because γ is a geodesic and $[\bar{x}, \bar{y}]$ is also a
 933 geodesic in $\mathbb{M}_K^2$, the pair $\gamma(t) \leftrightarrow \bar{\gamma}(t)$ correspond naturally for the sub-segment ratio t . Here, we
 934 have

$$d(p, \gamma(t)) \leq d_{\mathbb{M}_K^2}(\bar{p}, \bar{\gamma}(t)) + C_1\epsilon,$$

935 for some constant C_1 . By taking squares,

$$d^2(p, \gamma(t)) \leq \left(d_{\mathbb{M}_K^2}(\bar{p}, \bar{\gamma}(t))\right)^2 + 2C_1\epsilon d_{\mathbb{M}_K^2}(\bar{p}, \bar{\gamma}(t)) + (C_1\epsilon)^2.$$

936 Since $K < 0$, the model space $\mathbb{M}_K^2$ is either Euclidean or hyperbolic. In both cases, it is known that
 $\{\bar{\gamma}(t) \mid t \in [0, 1]\} \subset [\bar{x}, \bar{y}]$

937 which yields $\bar{\gamma}(t)$ satisfying the usual convexity of the squared distance in a non-positive curvature
 938 setting.

$$\left(d_{\mathbb{M}_K^2}(\bar{p}, \bar{\gamma}(t))\right)^2 \leq (1-t) \left(d_{\mathbb{M}_K^2}(\bar{p}, \bar{x})\right)^2 + t \left(d_{\mathbb{M}_K^2}(\bar{p}, \bar{y})\right)^2.$$

939 Therefore,

$$d_{\mathbb{M}_K^2}(\bar{p}, \bar{\gamma}(t))^2 \leq (1-t)d^2(p, x) + td^2(p, y),$$

940 and

$$\begin{aligned} d^2(p, \gamma(t)) &\leq (1-t)d^2(p, x) + td^2(p, y) + 2C_1\epsilon \left(d_{\mathbb{M}_K^2}(\bar{p}, \bar{\gamma}(t))\right) + (C_1\epsilon)^2 \\ &\leq (1-t)d^2(p, x) + td^2(p, y) + 2C_1\epsilon D' + (C_1\epsilon)^2 \\ &\leq (1-t)d^2(p, x) + td^2(p, y) + C_2\epsilon D, \end{aligned}$$

941 for some constant $C_2 > 0$, where D' is the diameter of the model space, and can be bounded by local
 942 diameter D . This can be written as

$$f(\gamma(t)) = d^2(p, \gamma(t)) \leq (1-t)f(\gamma(0)) + tf(\gamma(1)) + C_2\epsilon D,$$

943 and it exactly states the approximate geodesic convexity for $f(x) = d^2(p, x)$. □

944 **Corollary 1** (Approximate Uniqueness of Fréchet Mean). Under the same ϵ -approximate $\text{CAT}(K)$
 945 assumptions, consider the Fréchet functional

$$F(x) = \int_{\mathcal{M}} d^2(y, x) d\nu(y). \quad (18)$$

946 for a compactly supported probability measure ν . Then, one has the following.

- 947 • A minimizer of F exists for any $\epsilon > 0$.
- 948 • If ϵ is small, any two minimizers m_1 and m_2 must lie within a small neighborhood of each
 949 other:

$$d(m_1, m_2) \leq O(\sqrt{\epsilon}). \quad (19)$$

950 Hence, strict uniqueness is replaced by an ϵ -dependent bound.

951 **Proposition 7** (Local Existence and Uniqueness). Let $\mathcal{M}$ be a geodesic metric space that is $\text{CAT}(K)$
 952 (or ϵ -approximately $\text{CAT}(K)$ space) locally in a geodesic ball $B(p_0, R)$. That is, for any geodesic
 953 triangle fully contained in $B(p_0, R)$, the usual $\text{CAT}(K)$ (or approximate) triangle comparison
 954 property holds. Suppose ν is a probability measure on $\mathcal{M}$ whose support $\text{supp}(\nu)$ is contained in
 955 $B(p_0, R)$. Define the Fréchet functional

$$F(x) = \int_{\mathcal{M}} d^2(y, x) d\nu(y).$$

956 Then, one has the following.

- 957 • The function $F(x)$ attains its minimum at some $m \in B(p_0, R)$.
- 958 • If $K > 0$ but $\text{diam}(\text{supp}(\nu)) < \frac{\pi}{2\sqrt{K}}$, or if $K \leq 0$ (no diameter restriction), then m is
 959 unique within $B(p_0, R)$.

960 In other words, the Fréchet mean m exists in the local ball $B(p_0, R)$ and is unique when the (local)
 961 curvature constraints enforce strict geodesic convexity.

962 **Proposition 8** (Heavy-Tailed Distributions and Slower Convergence). Let $\mathcal{M}$ be either a strict
 963 $\text{CAT}(K)$ space or an ϵ -approximate $\text{CAT}(K)$ space of diameter $< D$. Suppose $Y_1, Y_2, \dots, Y_n$ are
 964 i.i.d. random points in $\mathcal{M}$ with common distribution ν . Denote by

$$\begin{aligned} \mu &= \arg \min_{z \in \mathcal{M}} \mathbb{E}[d^2(Y, z)] \\ \hat{\mu} &= \arg \min_{z \in \mathcal{M}} \frac{1}{n} \sum_{i=1}^n d^2(Y_i, z). \end{aligned}$$

965 Assume that

- 966 1. ν has finite second moments $\mathbb{E}[d^2(Y, z_0)] < \infty$ for some reference point z_0 , and
- 967 2. the random variable $d^2(Y, z_0)$ satisfies a sub-exponential-type tail bound: there exist
 968 constants $\alpha > 0, \gamma \in (0, 1]$ such that

$$\mathbb{P}(d^2(Y, z_0) > t) \leq \exp(-\alpha t^\gamma), \quad (20)$$

969 for all $t > 0$.

970 Then, there exist constants c, C such that for all $n > 1$ and all $\epsilon > 0$,

$$\mathbb{P}(d(\hat{\mu}_n, \mu) \geq \epsilon) \leq C \exp(-c n \epsilon^{2\gamma}). \quad (21)$$

971 Hence $\hat{\mu}_n$ converges to μ in probability, and its deviation tails decay sub-exponentially with rate $\epsilon^{2\gamma}$.

972 *Proof.* Define the population and empirical Fréchet functionals

$$F(z) = \mathbb{E}[d^2(Y, z)], \quad F_n(z) = \frac{1}{n} \sum_{i=1}^n d^2(Y_i, z).$$

973 By definition,

$$\mu = \arg \min_{z \in \mathcal{M}} F(z), \quad \hat{\mu}_n = \arg \min_{z \in \mathcal{M}} F_n(z).$$

974 Observe that

$$\begin{aligned} F(\hat{\mu}_n) - F(\mu) &= \{F(\hat{\mu}_n) - F_n(\hat{\mu}_n)\} + \{F_n(\hat{\mu}_n) - F_n(\mu)\} + \{F_n(\mu) - F(\mu)\} \\ &\leq \{F(\hat{\mu}_n) - F_n(\hat{\mu}_n)\} - \{F(\mu) - F_n(\mu)\}, \\ |F(\hat{\mu}_n) - F(\mu)| &\leq |F(\hat{\mu}_n) - F_n(\hat{\mu}_n)| + |F(\mu) - F_n(\mu)|. \end{aligned}$$

975 Therefore,

$$\{d(\hat{\mu}_n, \mu) \geq \epsilon\} \subseteq \{F(\hat{\mu}_n) - F(\mu) \geq \alpha(K, D)\epsilon^2\} \subseteq \left\{ \sup_{z \in \mathcal{M}} |F_n(z) - F(z)| \geq \frac{\alpha(K, D)}{2} \epsilon^2 \right\}.$$

976 Here,

$$\sup_{z \in \mathcal{M}} |F_n(z) - F(z)| \leq \max_{1 \leq j \leq N_\delta} |F_n(z_j) - F(z_j)| + \eta(\delta),$$

977 where $N_\delta \leq \exp(C_1(D/\delta)^m)$ is a δ -net for some m and $\eta(\delta) \rightarrow 0$ as $\delta \rightarrow 0$. Taking $\delta \rightarrow 0$,

$$\begin{aligned} \mathbb{P} \left(\sup_{z \in \mathcal{M}} |F_n(z) - F(z)| \geq t \right) &\leq N_\delta \cdot 2 \exp(-c' n t^\gamma) + \mathbb{P}(\eta(\delta) \geq t/2) \\ &\approx \exp(\ln N_\delta - c' n t^\gamma). \end{aligned}$$

978 For fixed D , $\log N_\delta$ is polynomial in $(1/\delta)$ so we can absorb that into a constant factor. □

979 D Relation to Geodesic Regression

980 A Riemannian manifold $(\mathcal{M}, g)$ is a smooth manifold endowed with a Riemannian metric g , which
981 locally induces a norm on each tangent space $T_p \mathcal{M}$. In such a setting, the geodesic distance between
982 two points $p, q \in \mathcal{M}$ is given by

$$d(p, q) = \inf_{\gamma} \int_0^1 \sqrt{g(\dot{\gamma}(t), \dot{\gamma}(t))} dt,$$

983 where the infimum is taken over all smooth curves γ joining p and q . For points q in a normal
984 neighborhood of p , the exponential map $\exp_p: T_p \mathcal{M} \rightarrow \mathcal{M}$ is a diffeomorphism and we have the
985 local relation

$$d^2(p, q) = \|\exp_p^{-1}(q)\|^2.$$

986 Moreover, assuming the sectional curvatures of $\mathcal{M}$ are bounded above by K , the manifold is
987 also a $\text{CAT}(K)$ space. In this smooth setting, one can use differential calculus; for example,
988 the Fréchet functional $F(z) = \int_{\mathcal{M}} d^2(y, z) d\nu(y)$ is differentiable (at least locally), with gradient
989 $\nabla F(z) = -2 \int_{\mathcal{M}} \exp_z^{-1}(y) d\nu(y)$, and a second-order expansion

$$F(\exp_z(v)) = F(z) + \langle \nabla F(z), v \rangle + \frac{1}{2} \langle H_z v, v \rangle + O(\|v\|^3).$$

990 Here, a $\text{CAT}(K)$ space is a geodesic metric space $(\mathcal{M}, d)$ satisfying a comparison condition: for
991 any geodesic triangle $\triangle pqr$ with perimeter less than a critical value (for $K > 0$) and any points x
992 and y on two of its sides, the distance $d(x, y)$ is bounded above by the corresponding distance in the
993 model space $\mathbb{M}_K^2$ of constant curvature K . In particular, if $\gamma: [0, 1] \rightarrow \mathcal{M}$ is a geodesic, one has the
994 following (strong) convexity inequality for the squared distance function:

$$d^2(y, \gamma(t)) \leq (1-t)d^2(y, \gamma(0)) + t d^2(y, \gamma(1)) - \alpha t(1-t) d^2(\gamma(0), \gamma(1)),$$

995 where $\alpha = \alpha(K, D)$ is a constant depending on the curvature bound K and the diameter D of
996 the region under consideration. The above inequality replaces the role of second-order (Hessian)
997 information.

998 In geodesic regression on a Riemannian manifold, we assume that the regression function follows a
 999 geodesic curve. For example, for a predictor $x \in \mathbb{R}^d$, one common formulation is:

$$\mu(x) = \exp_p((\alpha + \beta^\top x) v), \quad \text{with } v \in T_p \mathcal{M}.$$

1000 or equivalently, writing the geodesic γ from p with initial velocity v ,

$$\mu(x) = \gamma(\alpha + \beta^\top x).$$

1001 Here, $p \in \mathcal{M}$ is a base point, $v \in T_p \mathcal{M}$ is a tangent vector at p , $\exp_p$ is the Riemannian exponential
 1002 map, and α, β are the regression parameters. This model implies that the conditional mean of Y
 1003 given $X = x$ lies exactly on the geodesic determined by p and v . Fréchet regression is defined more
 1004 generally and does not restrict the mean to lie on a pre-specified geodesic. For each x , the conditional
 1005 Fréchet mean is given by

$$\mu(x) = \arg \min_{z \in \mathcal{M}} \mathbb{E} \left[d^2(Y, z) \mid X = x \right].$$

1006 If $\mathcal{M}$ is a Riemannian manifold and the conditional distribution of Y given $X = x$ is concentrated
 1007 and symmetric around a geodesic curve, then one may find that the minimizer satisfies

$$\mu(x) = \exp_p((\alpha + \beta^\top x) v),$$

1008 thus recovering the geodesic regression solution. However, in general, Fréchet regression allows
 1009 for much more flexible conditional mean structures. In summary, we can relate these two concepts
 1010 (Fréchet regression and geodesic regression). Riemannian manifolds allow a local linearization via
 1011 the exponential map and a full Taylor expansion, making geodesic regression a natural parametric
 1012 model, and $\text{CAT}(K)$ spaces provide a more general setting where one relies on strong convexity
 1013 properties of the squared distance function rather than differentiability. Both approaches are unified
 1014 under the Fréchet regression framework, with geodesic regression emerging as a parametric case
 1015 when the conditional means lie on a geodesic.

1016 E Details of Experiments

1017 This section describes the details of experiments in Section 4.

1018 **Model Details** Throughout the experiment, we use an implementation of Fréchet regression based
 1019 on the Nadaraya-Watson estimator [14, 21, 38].

$$\mu^*(x) = \arg \min_{z \in \mathcal{M}} \frac{1}{n} \sum_{i=1}^n K_h(X_i - x) d^2(Y_i, z),$$

1020 where K_h is a smoothing kernel that corresponds to a probability density with $K_h(\cdot) = h^{-1}K(\cdot/h)$.
 1021 For the optimization, we use Limited-memory BFGS [30].

```

1      import numpy as np
2      from scipy.optimize import minimize
3
4      # Kernel function (Gaussian kernel)
5      def gaussian_kernel(x, x_data, bandwidth):
6          dists = np.linalg.norm(x_data - x, axis=1)
7          weights = np.exp(-0.5 * (dists / bandwidth) ** 2)
8          return weights / np.sum(weights)
9
10     # Fréchet objective function
11     def frechet_objective(y, responses, weights, distance_func):
12         dists = np.array([distance_func(y, r) for r in responses])
13         return np.sum(weights * dists**2)
14
15     # Fréchet regression function
16     def frechet_regression(X, Y, x_query, bandwidth, distance_func):
17         weights = gaussian_kernel(x_query, X, bandwidth)
18         y_init = np.mean(Y, axis=0)
19         result = minimize(
20             frechet_objective,
21             y_init,
22             args=(Y, weights, distance_func),
23             method='L-BFGS-B'
24         )
25         return result.x

```

Listing 1: Python code for the Fréchet regression.

1022 **Stereographic Projection** Listing 2 shows the Python code for the stereographic projection from
 1023 sphere surface to hyperbolic plane.

```

1      # Define the stereographic projection function
2      def stereographic_projection(x, y, z, R):
3          u = R * x / (R + z)
4          v = R * y / (R + z)
5      return u, v

```

Listing 2: Python code for the stereographic projection.

1024 E.1 Details for Illustrative Example 4.1

1025 **Data Generating Process** To assess the performance of the Fréchet regression estimator, consider
1026 to generate simulated data. The regression function is

$$\mu(x)(\cdot) = ((1 - x^2)^{1/2} \cos(\pi x), (1 - x^2)^{1/2} \sin(\pi x), x), \quad x \in (0, 1),$$

1027 which maps a spiral on the sphere. To generate a random sample $\{(X_i, Y_i)\}_{i=1}^n$, let $X_i \sim \mathcal{U}(0, 1)$
1028 followed by a bivariate normal random vector U_i , and

$$Y_i = \cos(\|U_i\|)\mu(X_i) + \sin(\|U_i\|)\frac{U_i}{\|U_i\|}.$$

1029 The sample size of the simulation data is $n = 50$, and Gaussian noise with variance 0.4 is added to
1030 each instance.

1031 E.2 Details for Experiments on Real-world Datasets 4.2

1032 Details of Datasets

- 1033 • **HYG Stellar:** The HYG Stellar Database is a comprehensive star catalog that amalgamates
1034 data from several prominent astronomical catalogs, including HIPPARCOS, the Yale Bright
1035 Star Catalog, and the Gliese Catalog of Nearby Stars. This integration provides detailed
1036 information on stars' positions, brightness, spectral types, and various identifiers such as
1037 traditional names and Bayer designations. It contains detailed information on 119,614 stars
1038 including position data, photometric data and luminosity and variability.
- 1039 • **USGS Earthquake:** The USGS Earthquake catalogue provides information on earthquakes
1040 worldwide with a magnitude of 2.5 and above that have occurred over the past week, and it
1041 contains 300 instances.
- 1042 • **NOAA Climate:** The NOAA Climate data provides Two-Line Element (TLE) sets for
1043 weather satellites, including those operated by NOAA, and contains 72 instances. A TLE
1044 consists of two 69-character lines of data, each containing specific parameters that describe
1045 the satellite's orbit.

1046 Table 3 shows the detailed breakdown of variables X and Y for each dataset.

Dataset	Sample size	Predictor X	Response Y
HYG Stellar	119,614	• Observation time t • Brightness of the star m • Absolute Magnitude m' • Spectral type s	Position on the celestial sphere
USGS Earthquake	300	• Observation time t • Magnitude of the earthquake m • Depth of the earthquake d	Earthquake location
NOAA Climate	72	• Timestamp of the TLE t • Orbital parameters θ • Inclination i	Satellite position

Table 3: Detailed breakdown of variables for each dataset.

1047 **Visualizations of Real-world Spherical Datasets** Figure 5 shows the additional visualizations of
1048 real-world spherical datasets, and Figure 6 shows the heteroscedasticity in the NOAA and USGS
1049 datasets. In addition, Python code in Listing 3 shows the implementation for the visualization of
1050 HYG Stellar dataset.

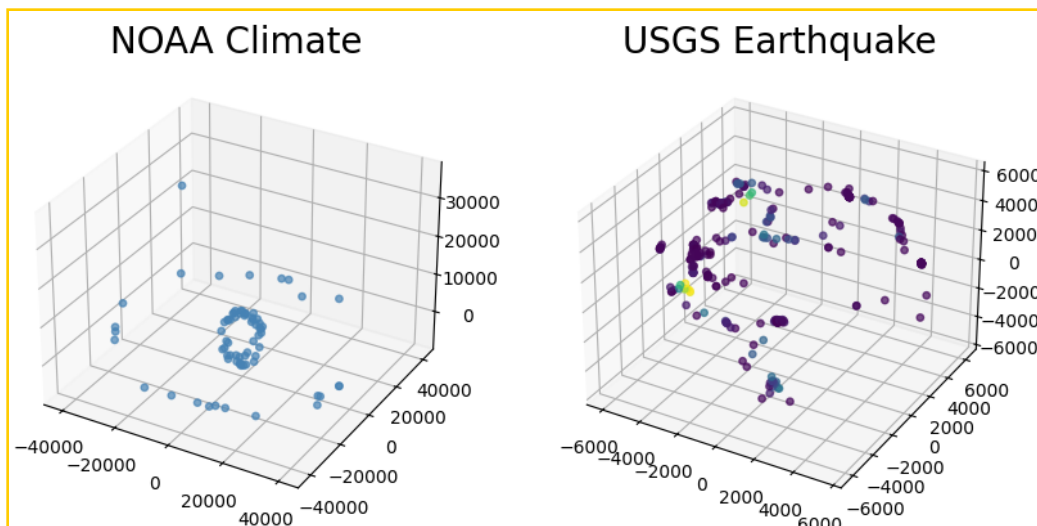


Figure 5: Visualizations for USGS Earthquake catalogue and NOAA Climate dataset.

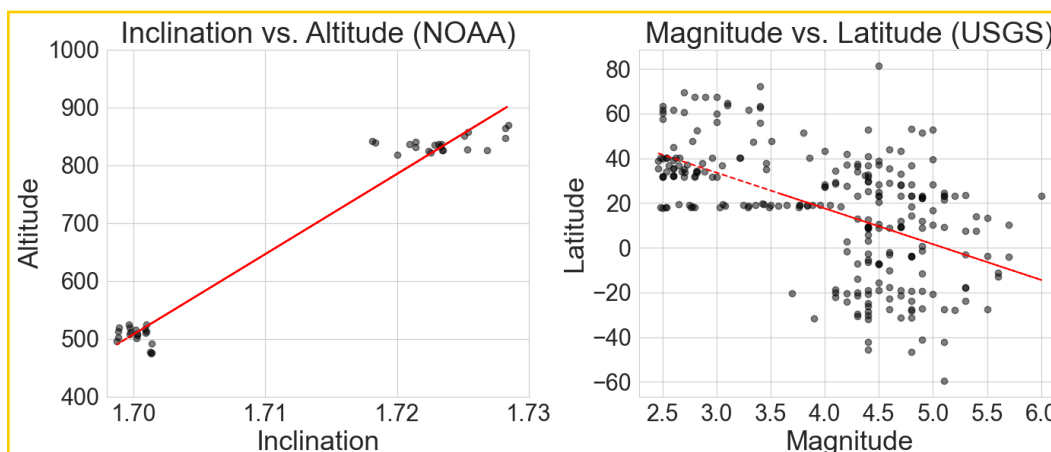


Figure 6: Heteroscedasticity in the NOAA and USGS datasets.

```

1
2 import numpy as np
3 import matplotlib.pyplot as plt
4 from astropy.io import ascii
5
6 # Load the Bright Star Catalog
7 url = '{Data URL}' # URL for HYG Steller database
8 data = ascii.read(url)
9
10 # Extract Right Ascension and Declination
11 ra = np.array(data['ra']) # in hours
12 dec = np.array(data['dec']) # in degrees
13
14 # Convert RA from hours to degrees
15 ra_deg = ra * 15
16
17 # Convert RA and Dec to radians for plotting
18 ra_rad = np.radians(ra_deg)
19 dec_rad = np.radians(dec)
20
21
22 # Create a 3D scatter plot
23 fig = plt.figure(figsize=(12, 8))
24 ax = fig.add_subplot(111, projection='3d')
25
26 # Convert spherical coordinates to Cartesian for plotting
27 x = np.cos(dec_rad) * np.cos(ra_rad)
28 y = np.cos(dec_rad) * np.sin(ra_rad)
29 z = np.sin(dec_rad)
30
31 # Plot the stars
32 ax.scatter(x, y, z, color='white', s=0.01, label="data points")
33
34 ax.xaxis.set_ticklabels([])
35 ax.yaxis.set_ticklabels([])
36 ax.zaxis.set_ticklabels([])
37
38 # Set plot parameters
39 ax.set_facecolor('black')
40 ax.set_xlabel('X')
41 ax.set_ylabel('Y')
42 ax.set_zlabel('Z')
43 plt.legend(markerscale=80, fontsize=30)
44 plt.show()

```

Listing 3: Python code for the visualization of HYG Steller database.

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